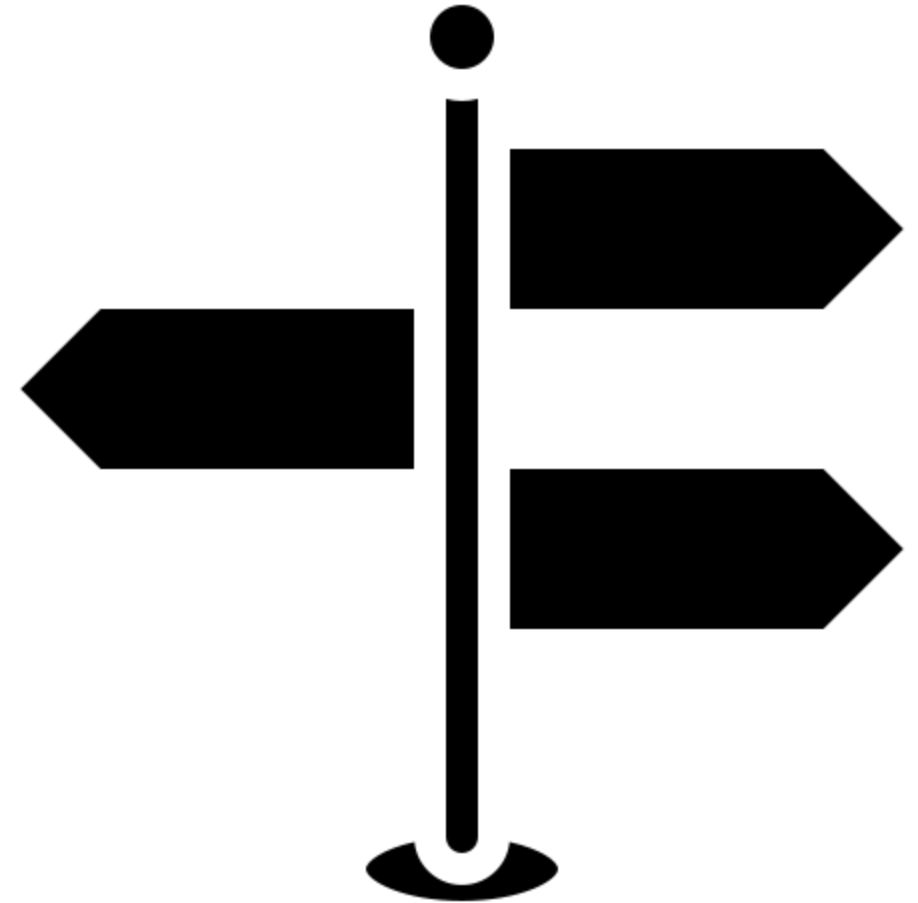


Public Cloud Services Network



Learning Objectives

- Students are familiar with the challenges of connecting the cloud to existing networks.
- Students can independently create and configure virtual networks in the cloud.
- Students are familiar with network services such as virtual networks and load balancer types.



Agenda

- Virtual Networks / Virtual Private Clouds (VPCs)
- Gateways & NAT
- Connectivity (VPN, Direct Connect/Express Route)
- DNS
- Service Integration
- Load Balancing

Motivation



Cloud Connectivity

- A very good Internet connection is essential for using public cloud services
 - Bandwidth and latency
 - Redundancy and availability
 - Choice of provider and transmission technology
- The customer's location often limits availability and bandwidth of the Internet connection (e.g., a major problem in Germany and France)
- All data is transmitted via the Internet
- The security of cloud services is based on encryption technologies



Cloud Connectivity – Challenges

- The use of cloud services leads to the following connectivity challenges:
 - Private networking on a multi-tenant infrastructure
 - Stable connection minimum bandwidth >1Gbit/s, latency <1ms
 - Redundancy if possible (active/active, active/passive)
 - Fiber transmission technology
 - Encryption of data in transit

Case Study: What is the latency of cloud access?

- How much latency is there when accessing cloud infrastructures in
 - Amsterdam
 - Auckland
 - Chennai
 - Dublin
 - Frankfurt
 - London
 - Los Angeles
 - Moscow
 - New York
 - ?

Case Study: How much latency is there when accessing cloud infrastructures in

Formula:

Latency = Lightspeed * 2/3 + x

(x = router overhead, actual distance traveled ...)

https://en.wikipedia.org/wiki/Velocity_factor

Conclusion:

Most latency today is physically limiting and cannot be reduced.

		Zurich
Amsterdam	✕	● 17.523ms
Auckland	✕	● 284.214ms
Chennai	✕	● 157.473ms
Dublin	✕	● 36.372ms
Frankfurt	✕	● 6.678ms
London	✕	● 25.934ms
Los Angeles	✕	● 157.37ms
Moscow	✕	● 45.17ms
New York	✕	● 96.014ms
Zurich	✕	—

Private networks in the cloud

Networking in the Cloud: Virtual Networks & Subnets

- Virtual networks are isolated, logical networks in the cloud. Underlying networking infrastructure is shared, but isolated for multiple tenants.
- Based on private IP addresses (e.g., IPv4 10.0.0.0/8, 172.16.0.0/12, or 192.168.0.0/16) or IPv6
- Each cloud customer can create multiple networks (possibly with the same private IP addresses) that are logically isolated from each other
- Services: Azure VNET / Virtual Network and AWS Virtual Private Cloud (VPC)



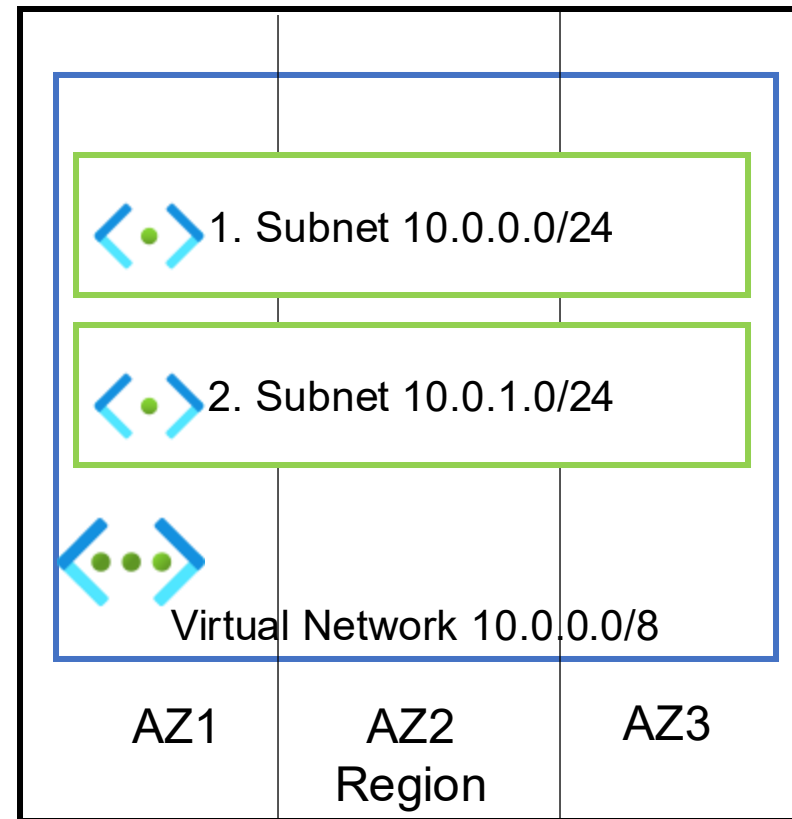
Azure Virtual Network



AWS Virtual Private Cloud

Virtual Networks & Subnets – Azure VNETs

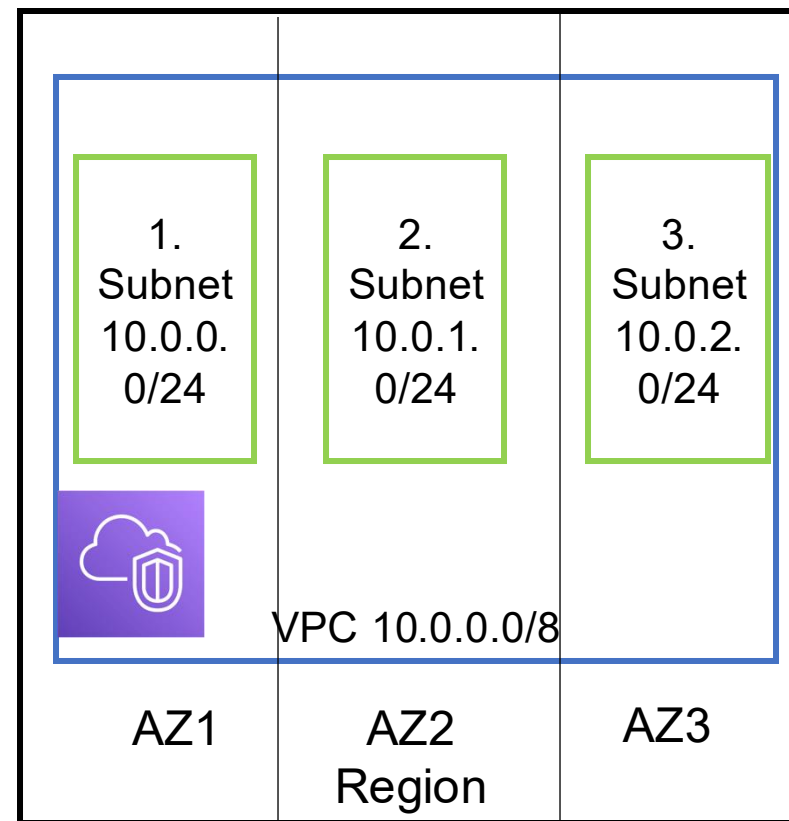
- Azure Virtual Networks are regional and apply to all Availability Zones. There is no default VNET.
- The VNET defines the CIDRs, e.g., 10.0.0.0/8.
- Subnets are assigned ranges, e.g., 10.0.0.0/24.
- Each subnet occupies at least 5 IPs, e.g. for a /24
 - .0 network address
 - .1 default gateway
 - .2 and .3 for Azure DNS
 - .255 broadcast address
- VNETs can be peered within the same region or between regions.



<https://learn.microsoft.com/en-us/azure/virtual-network/virtual-networks-faq>

Virtual Networks & Subnets – AWS VPCs

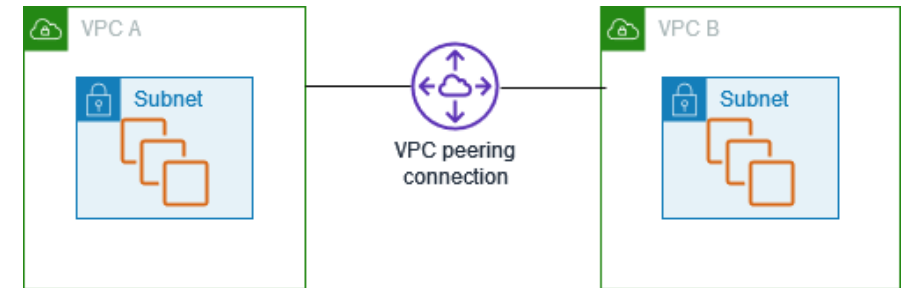
- AWS VPCs are regional, with one default VPC per region.
- The VPC defines the CIDRs, e.g., 10.0.0.0/8.
- Subnets are assigned ranges, e.g., 10.0.0.0/24. **Subnets are availability zone-specific.**
- Each subnet occupies at least 5 IPs, e.g., for a /24.
 - .0 network address
 - .1 VPC default gateway
 - .2 for DNS, .3 is reserved
 - .255 broadcast address
- VPCs can be peered within the same region or between regions.



<https://docs.aws.amazon.com/vpc/>

The need for network peerings

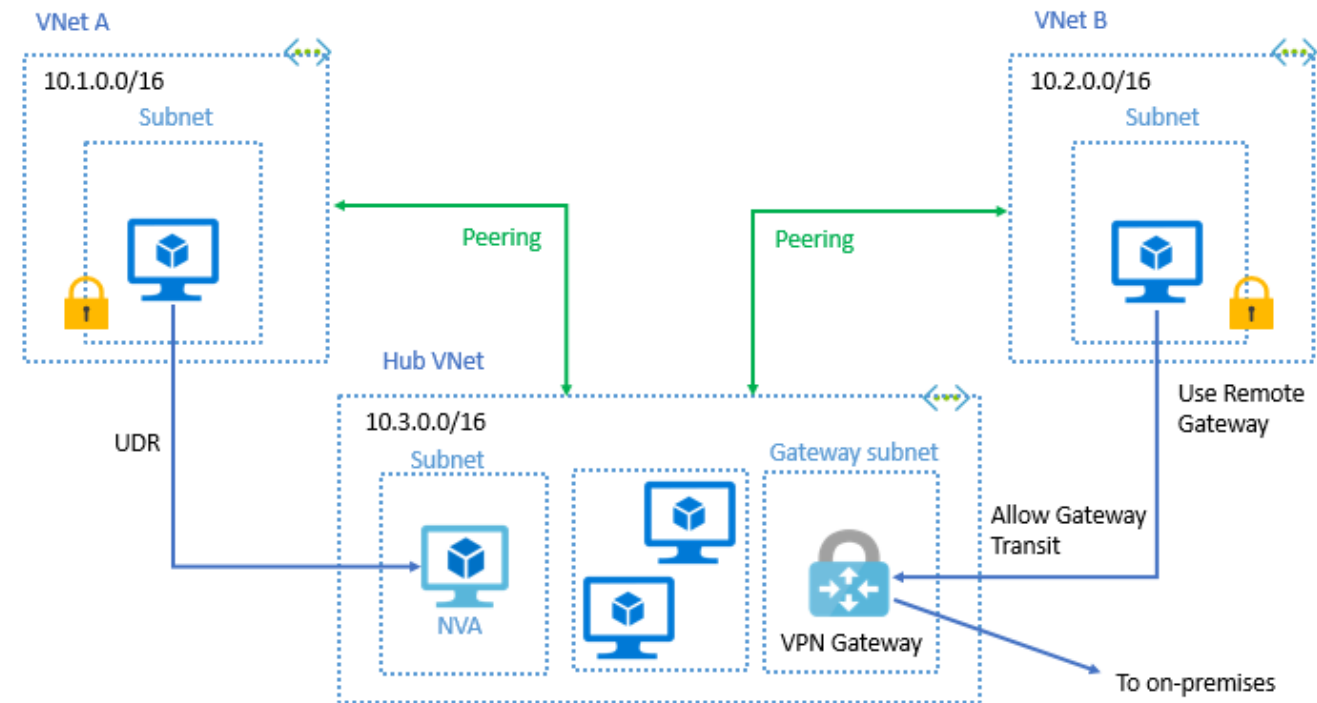
- Peerings are a native connections between VNETs / VPCs
- Services need to communicate privately
 - VPCs may reside in different accounts
 - VNETs may reside in different subscriptions
- Provisioning of VPNs (or others) to connect VNETs / VPCs is overhead
- Peerings are not free!
 - In- and outbound traffic volume is charged
 - Prices vary between cross AZ, region or even continents



<https://docs.aws.amazon.com/vpc/latest/peering/what-is-vpc-peering.html>

Network peerings (2)

- Architecture patterns such as hub-and-spoke force the use of multiple VNETs.
- Peerings link VNETs/VPCs with each other and enable IP-based communication.
- Use case: Increased autonomy for application teams and the central network team.



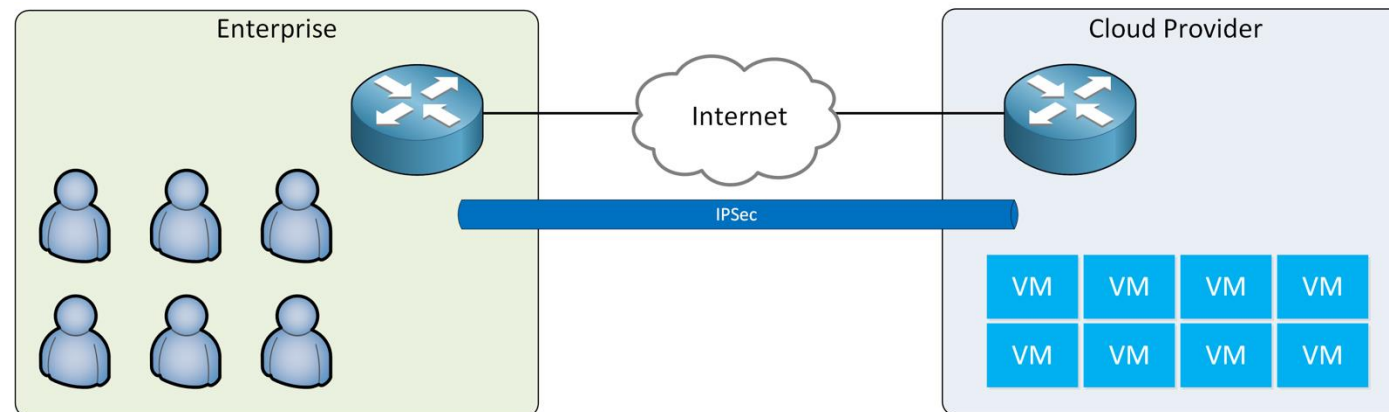
Peerings only work in the same cloud with the same technology (VNET <-> VNET or VPC <-> VPC)!

<https://learn.microsoft.com/de-de/azure/virtual-network/virtual-network-peering-overview>

Connect to private networks in the cloud

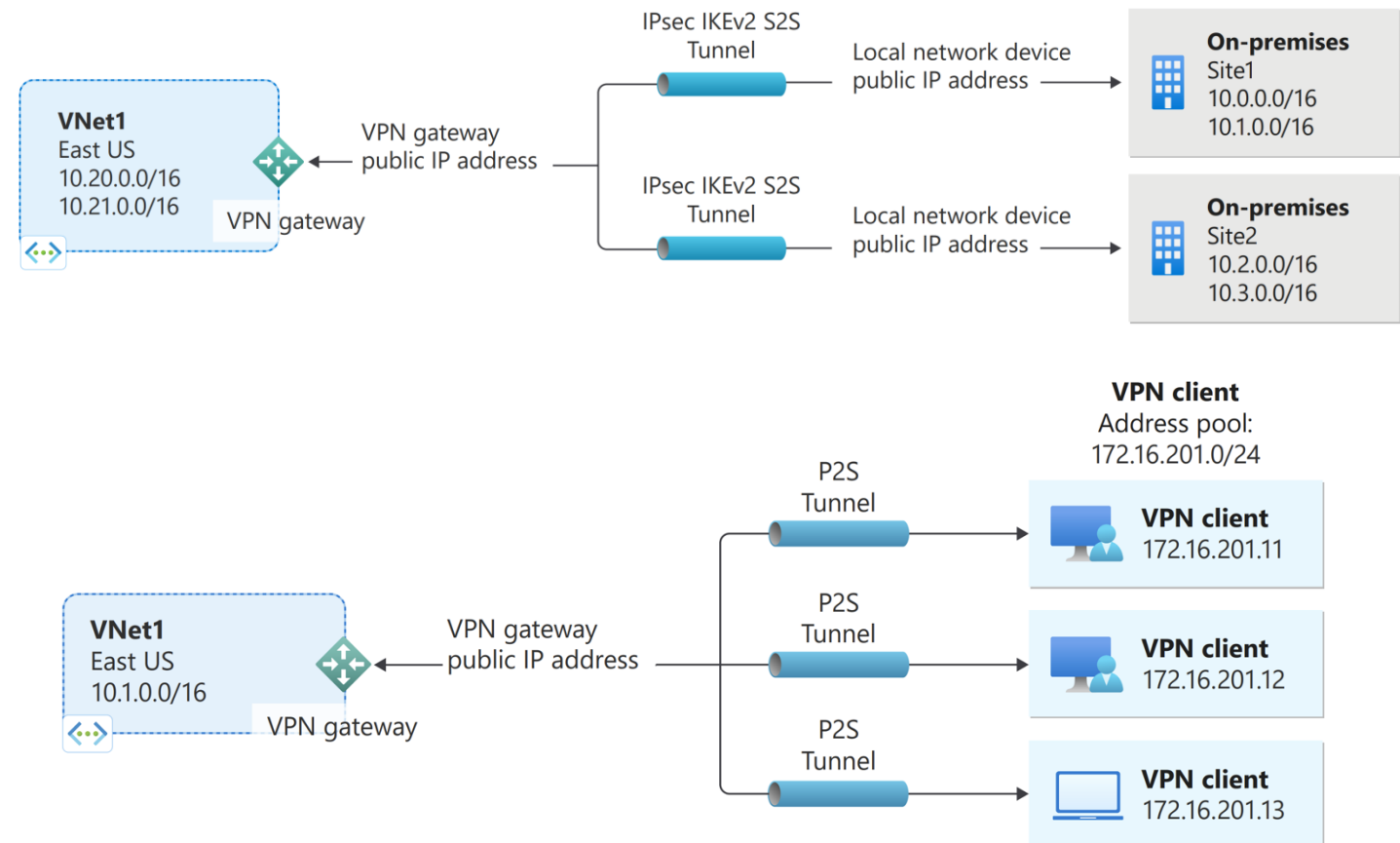
Virtual Private Network – VPN

- Point-to-site (P2S) or site-to-site (S2S) connection to your own VNET/VPC from a client or network
- Private communication with cloud resources via encrypted tunnels (e.g., IPsec) and internal IP addresses



Virtual Private Network – VPN – Azure VPN Gateway

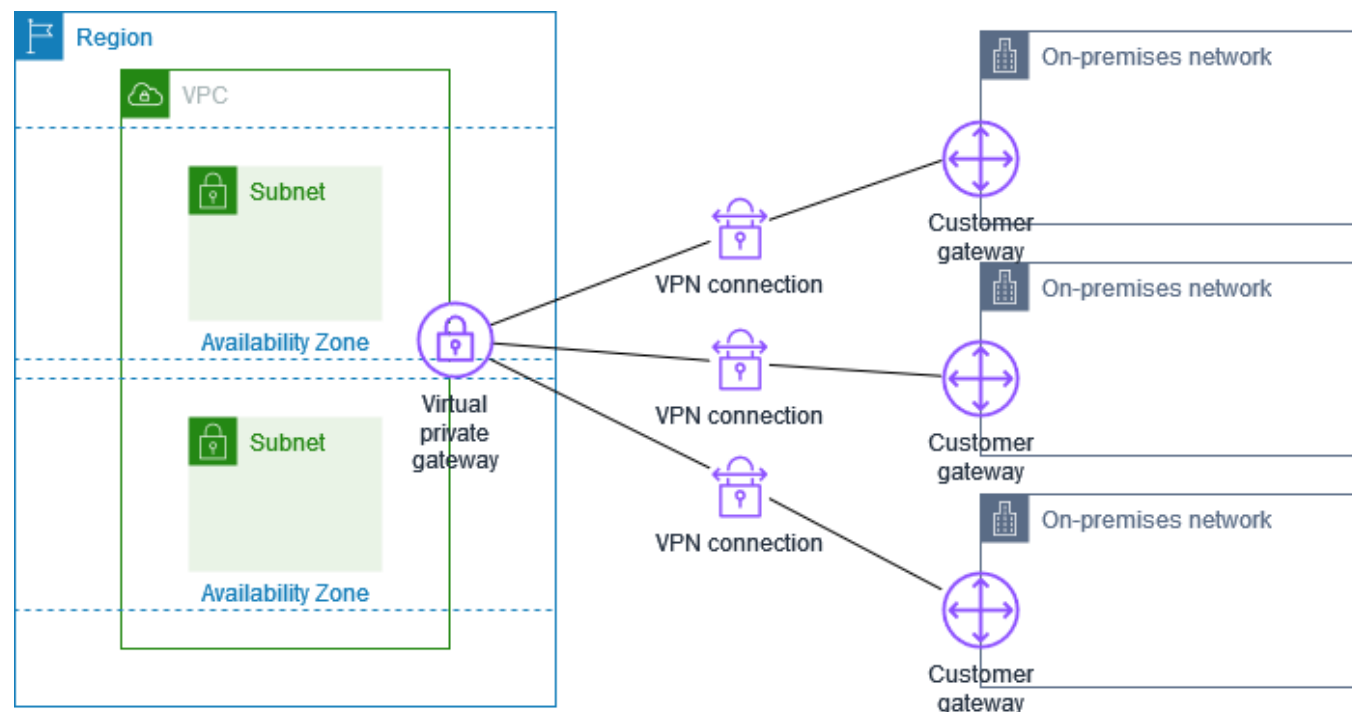
- One gateway can be created per VNET, e.g., in the form of a VPN gateway
- VPN gateway resource includes P2S and S2S connections
- SKU determines bandwidth
- Static and dynamic routing possible



<https://learn.microsoft.com/en-us/azure/vpn-gateway/design>

Virtual Private Network – VPN – AWS Virtual Private Gateway

- AWS Virtual Private Gateway is created as part of the VPC.
- Virtual Private Gateway contains S2S connections.
- Alternative: Transit Gateway – here, multiple VPCs are connected to the Transit Gateway. Dedicated routes in VPCs are required.



<https://docs.aws.amazon.com/vpn/latest/s2svpn/Examples.html>

VPN vs. dedicated connections

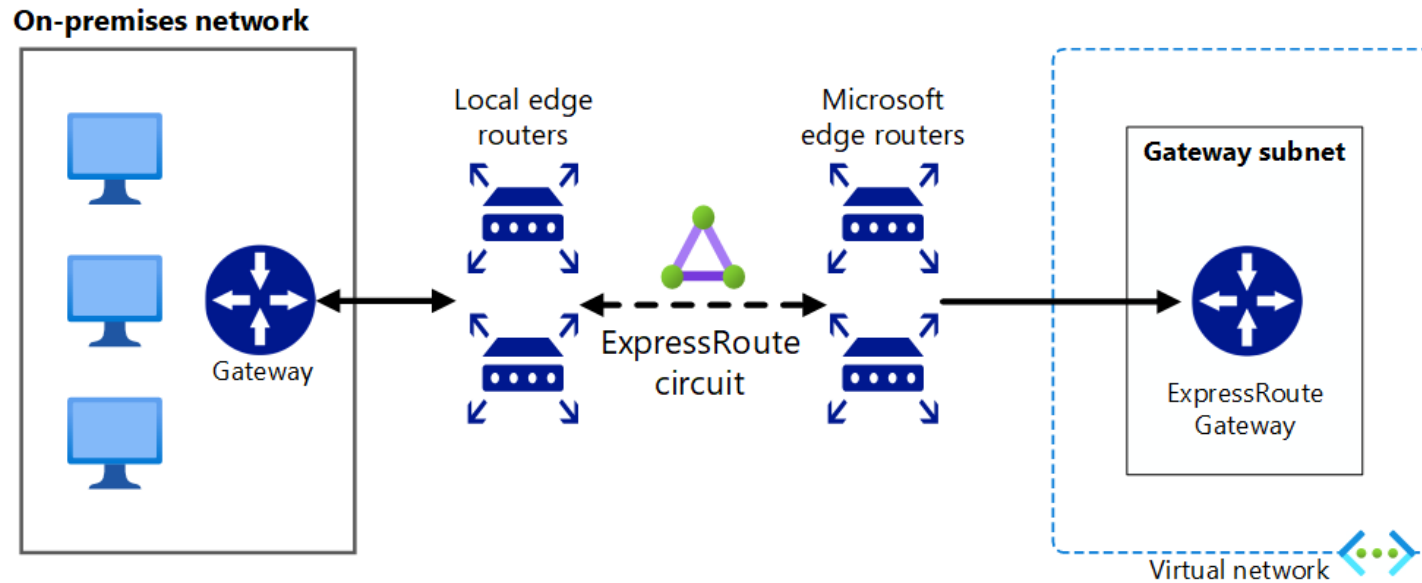
VPN:

- Quick to set up
- No lead time
- BYOIP
- Costs per time and data
- Limited bandwidth

Dedicated Connections:

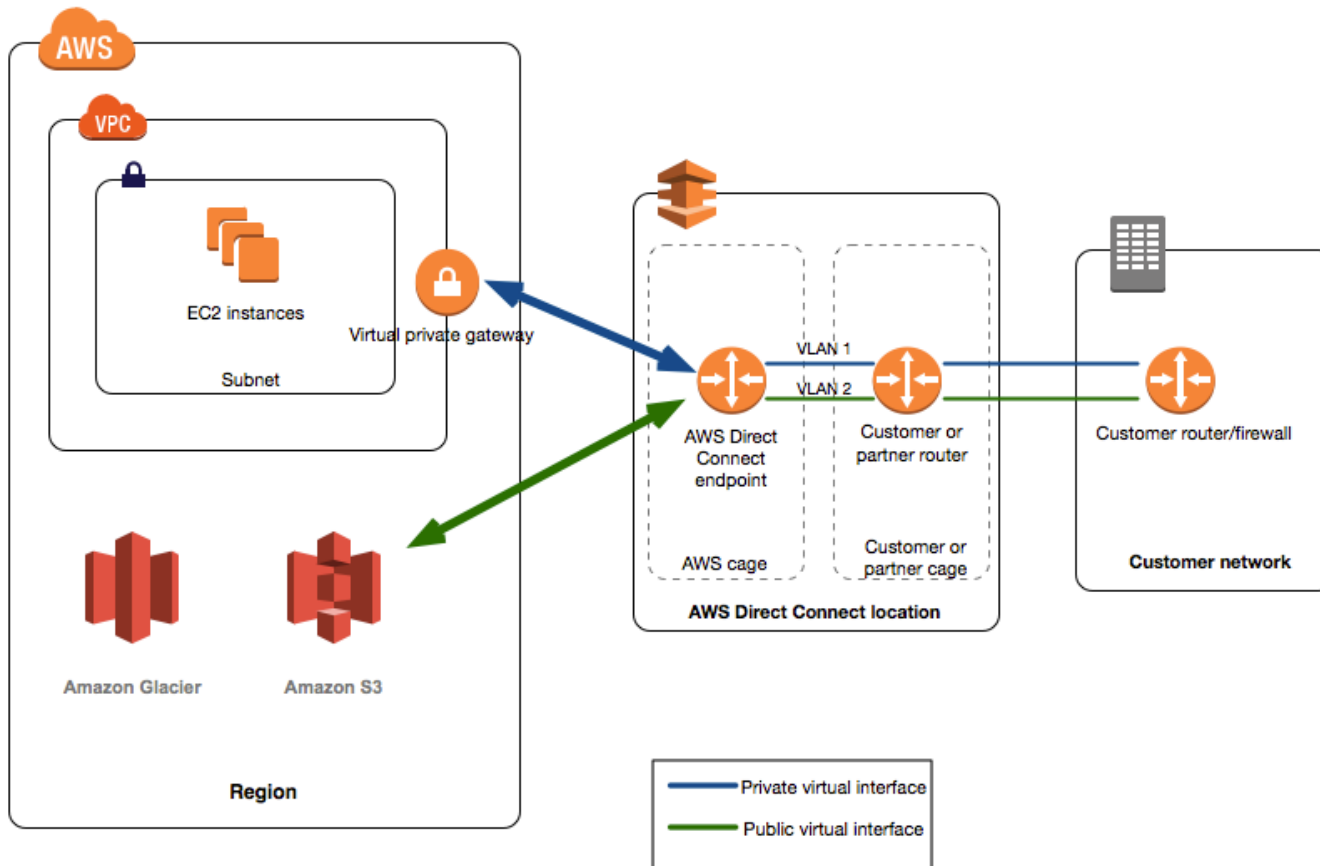
- Longer lead time, dedicated line
- Several weeks
- IP range by cloud
- Cost per data
- >100GBit possible

Azure Express Route



- Dedicated connection from the customer location (data center, campus) directly to Azure
- Guaranteed bandwidths
- Costs per traffic and bandwidth guarantee
- Low latency
- Service providers offer direct connections to Azure (peering).

AWS Direct Connect

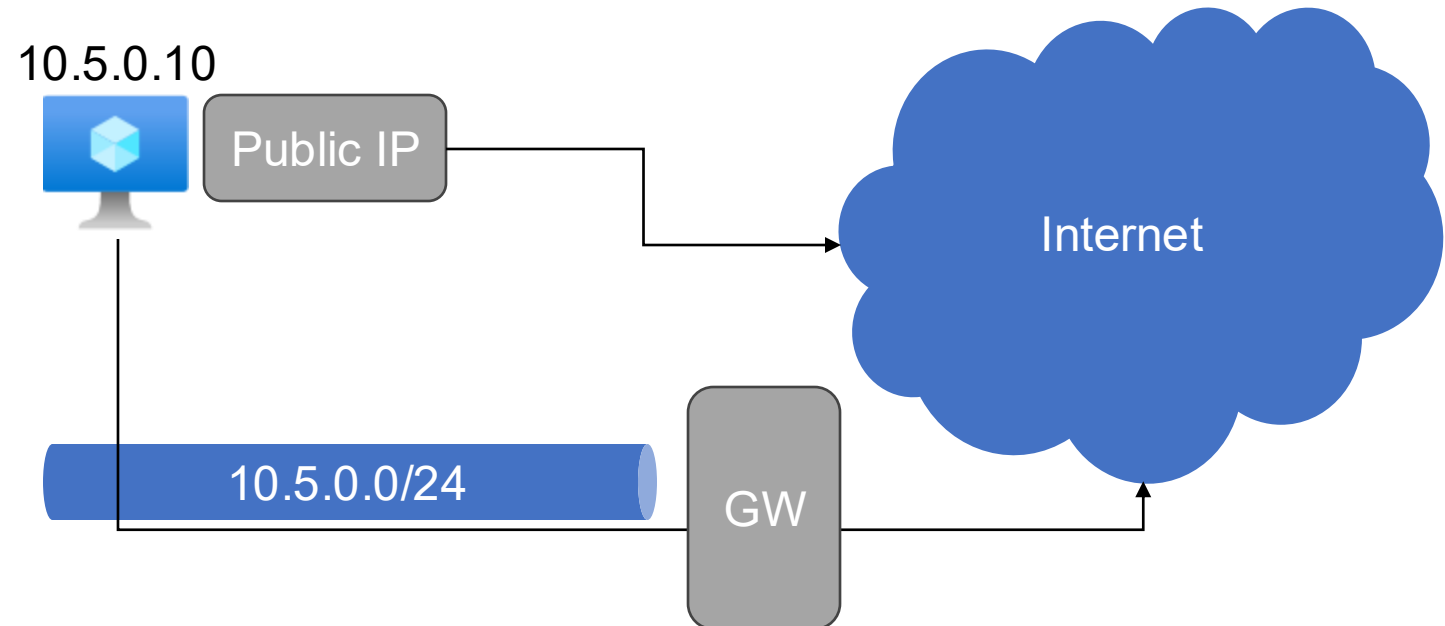


- Dedicated connection from the customer location (data center, campus) directly to the AWS cloud.
- Guaranteed bandwidths
- Costs per traffic and bandwidth guarantee
- Low latency
- Service providers offer direct connections to the AWS cloud (peering).

Management of outbound traffic

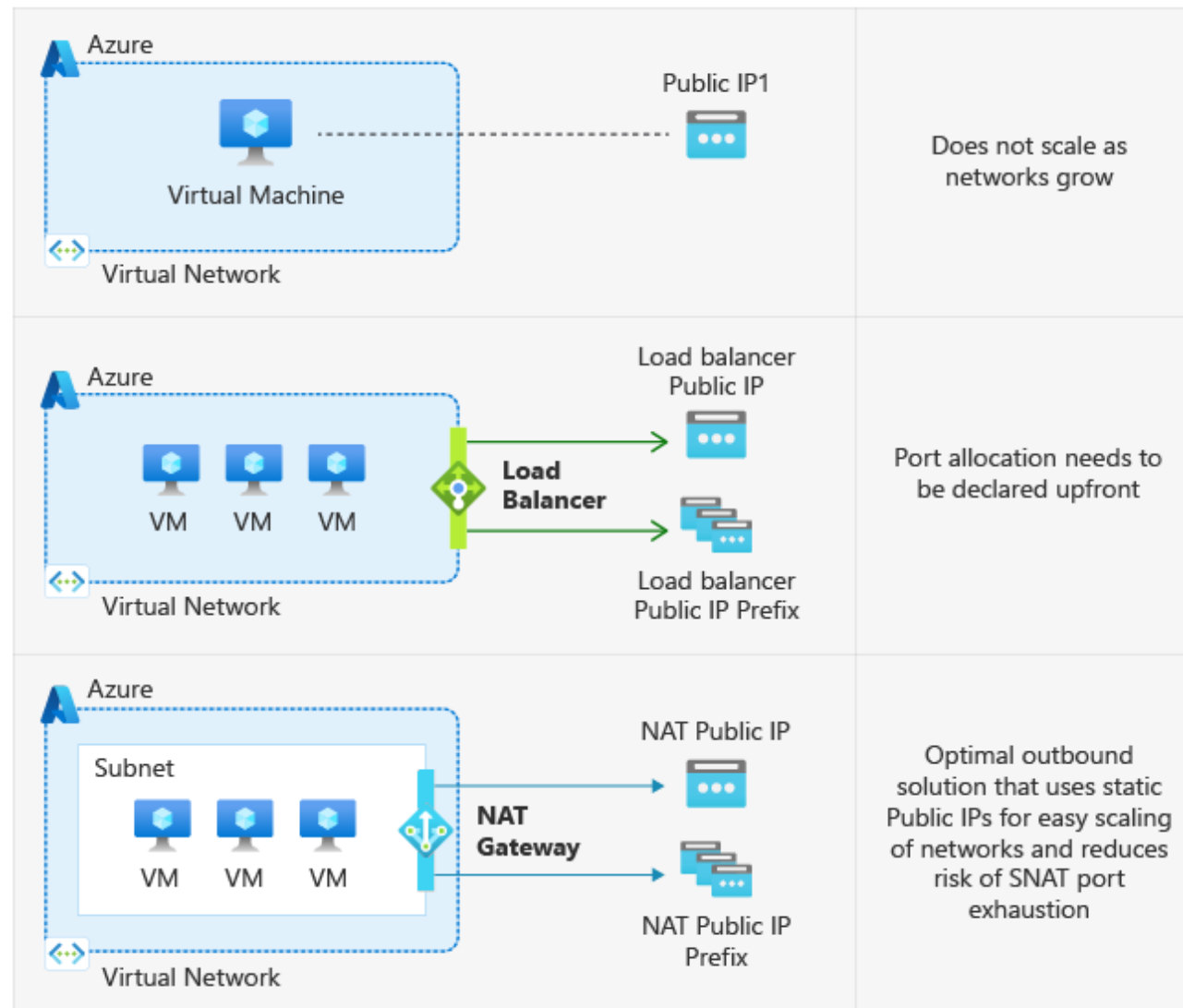
Gateways, NAT Gateway

- IPv4 requires NAT or similar to call an external service from VPC/VNET
- Challenges
 - Scalable solution for enterprises
 - Control over egress traffic (source/destination & content)
- Egress proxy installations are rare in the cloud; alternatives such as network firewalls & filters are used



Gateways, NAT Gateway – Azure

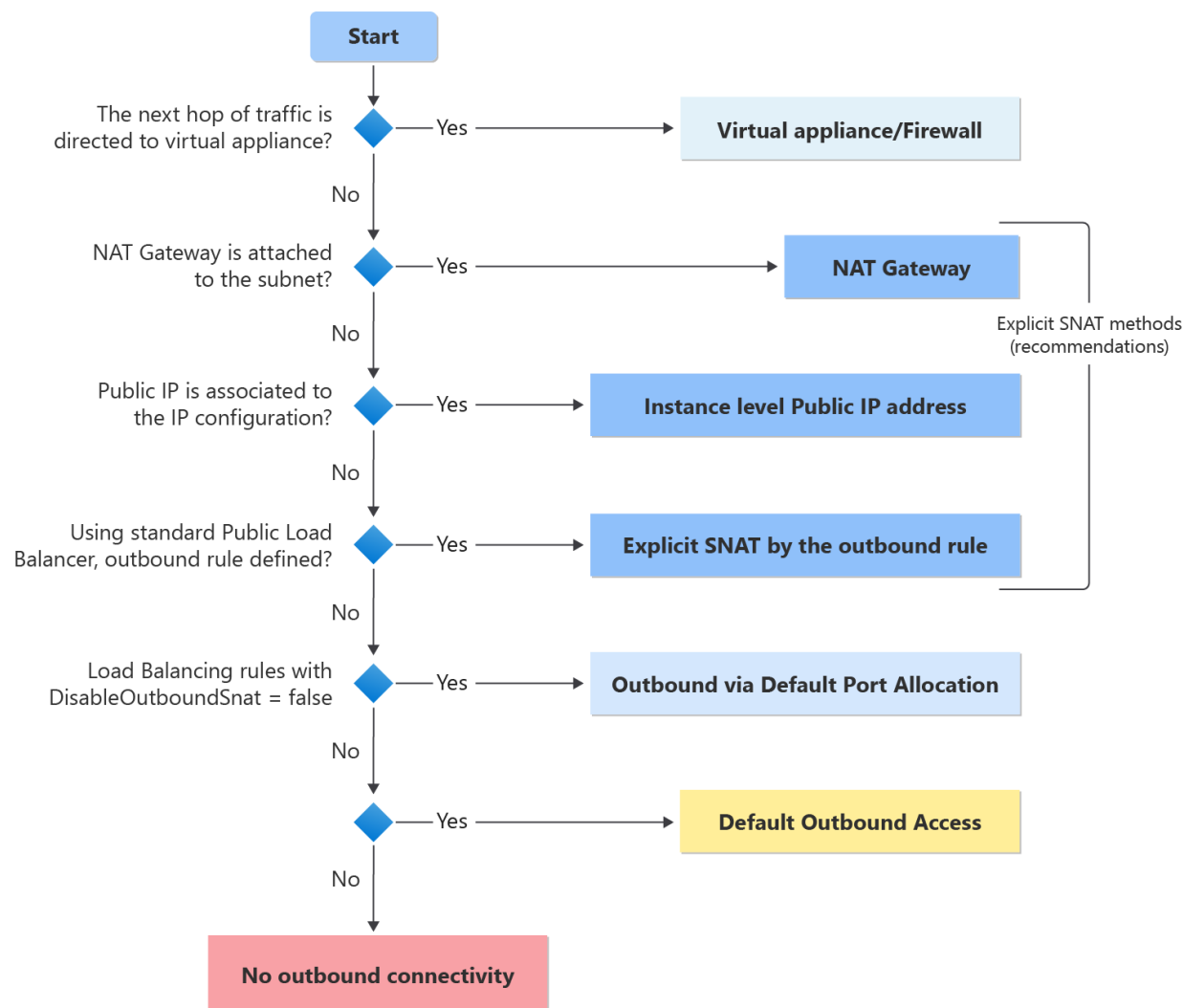
- Until March 31, 2026, Azure VNETs will allow standard outbound traffic via an Azure-managed IP (non-static).
- Egress traffic can be controlled using the following services
 - Dedicated public IPs (e.g., on a VM)
 - Load balancer IP (outbound SNAT)
 - NAT gateway
 - Virtual appliance/firewall



<https://learn.microsoft.com/en-us/azure/virtual-network/ip-services/default-outbound-access>

Gateways, NAT Gateway – Azure

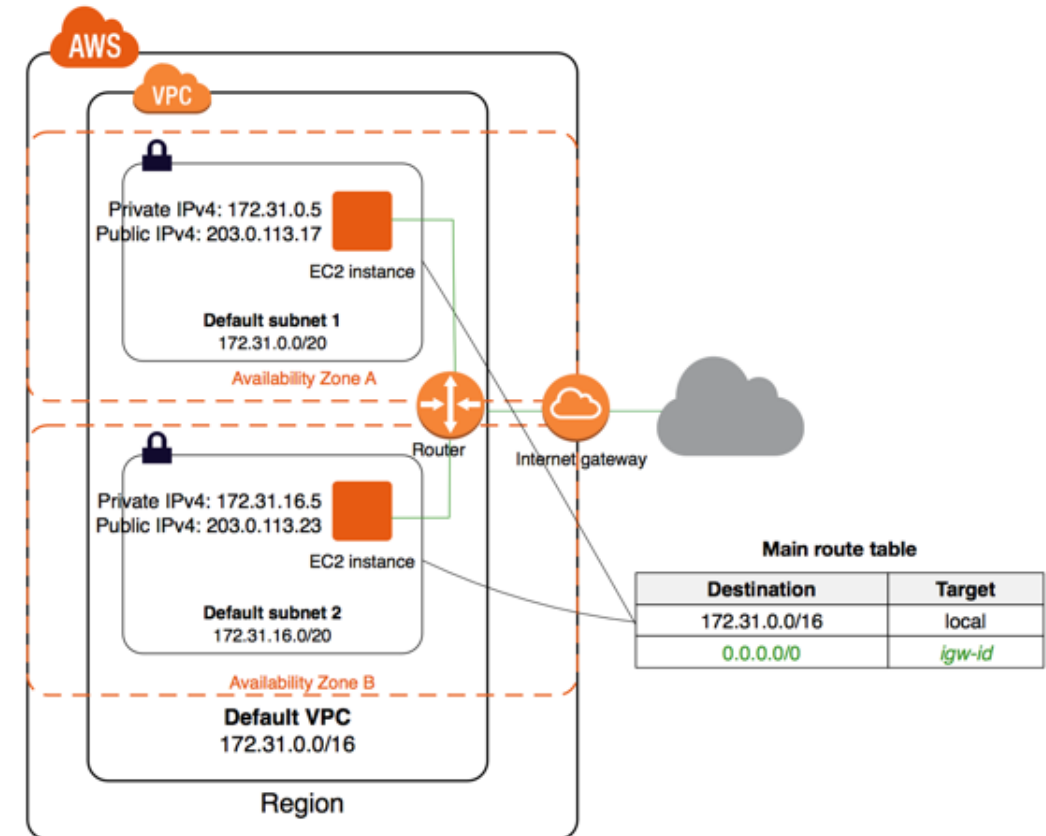
- Always define egress traffic for productive applications yourself
- Egress options have different priorities (see Flow)



<https://learn.microsoft.com/en-us/azure/virtual-network/ip-services/default-outbound-access>

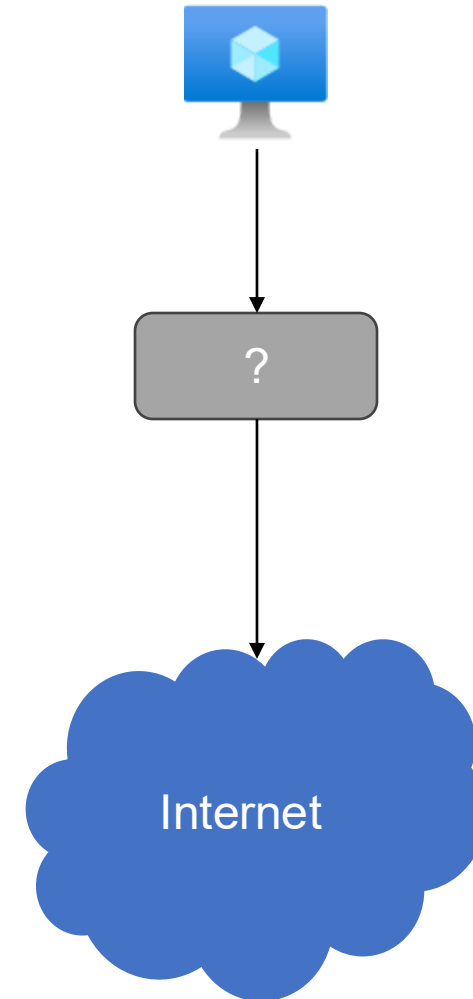
Gateways, NAT Gateway – AWS

- The AWS VPC does not allow egress traffic by default. It must always be explicitly defined, e.g., by assigning an IP address or using an Internet gateway.



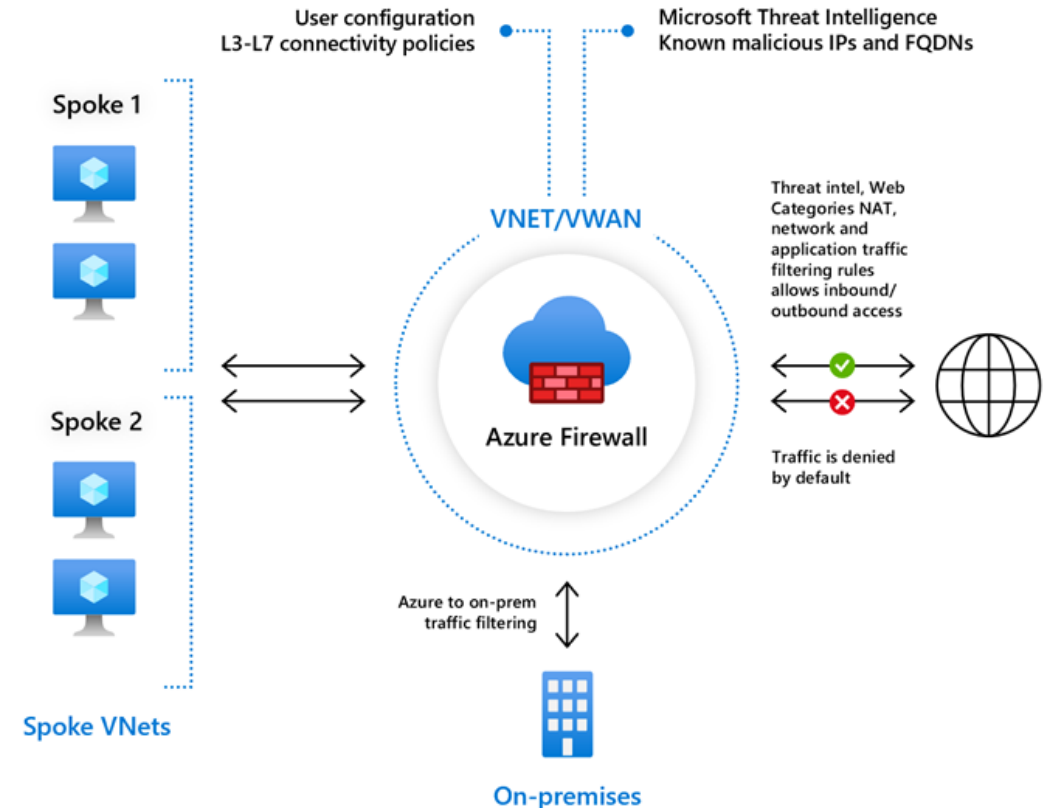
Gateways, NAT Gateway – Egress Traffic Filtering

- On-premises networks often use proxy installations for egress traffic filtering
- Clouds offer many options for establishing connections from VPCs/VNETs to the internet (public IPs, gateways, etc.)
- Centralized control and management by corporate IT is still necessary, even in the cloud



Gateways, NAT Gateway – Azure Firewall

- Azure Firewall is an L3/4 managed firewall service with features such as
 - FQDN filtering
 - Network filtering
 - DNS proxy
 - IDPS
 - HA via availability zones
- Centrally deployed and as the primary node for egress traffic (via 0.0.0.0/0 route)



<https://learn.microsoft.com/en-us/azure/firewall/features-by-sku>

Gateways, NAT Gateway – AWS Network Firewall

- Managed firewall solution for protecting VPCs
- Traffic protection and monitoring
- Threat detection and blocking
- Perimeter security and access control



DNS in the cloud

Recap: What is DNS?

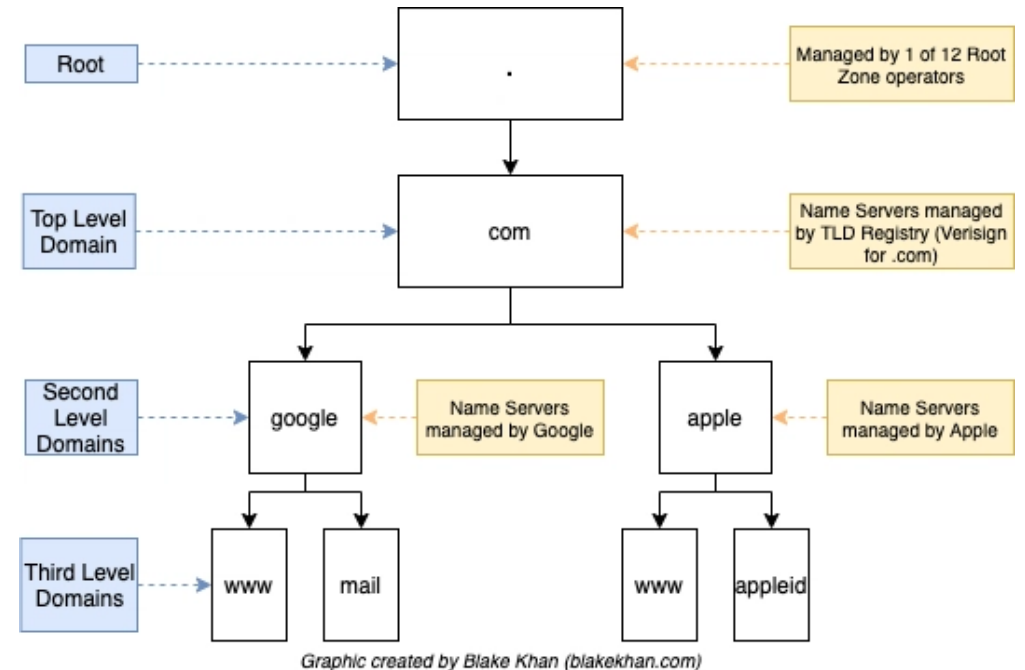
- DNS = Domain Name System
- Defined in [RFC 1035](#)

“The goal of domain names is to provide a mechanism for naming resources in such a way that the names are usable in different hosts, networks, protocol families, internets, and administrative organizations.” (2.1. Overview)

➔ Effectively a translation from human readable domains like <https://fhnw.ch> to IPs or other names.

Recap: What is DNS? - Hierarchy

- Domains are by design a hierarchy
- Each level may be operated by a different entity (represented by different DNS servers)
- Delegation of (sub) domains work with NS records
- DNS servers typically work on port 53 (TCP or UDP)
- DNS records (see next slide) have a TTL (time to live)



<https://dev.to/blake/dns-explained-hierarchy-and-architecture-18pj>

Recap: What is DNS? - Common RRs (Resource Records)

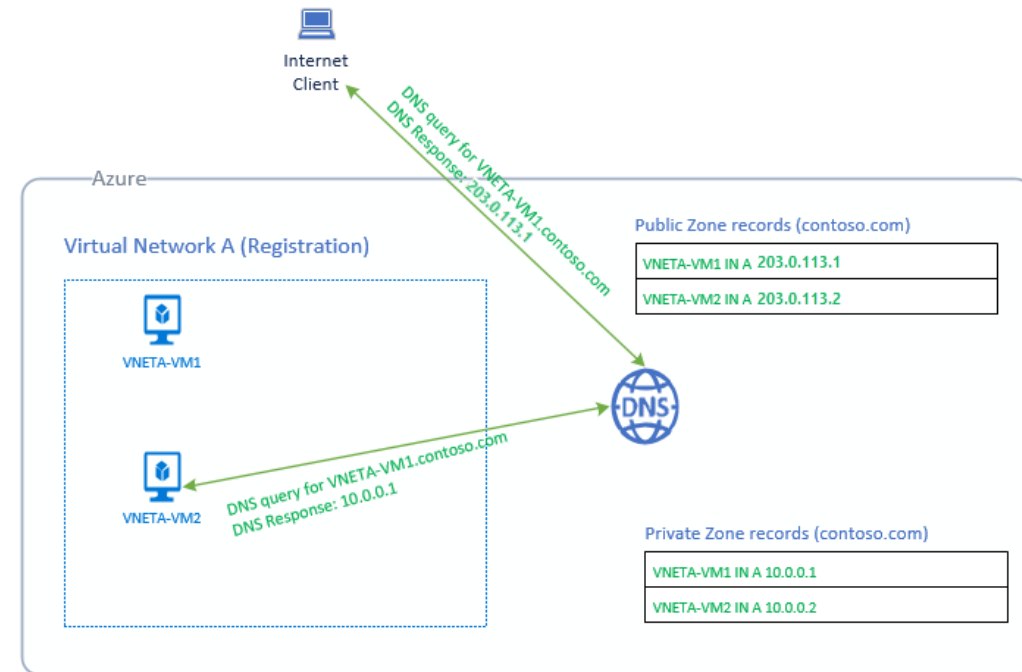
Resource Record	Example	Use Case
A-Record (or AAAA-Record for IPv6)	mydomain.ch -> 1.2.3.4	Pointer from a domain name to an IP address.
CNAME	mydomain.ch -> web.yourdomain.ch	Creation of an alias record for an existing domain, without tracking its “real” IP.
TXT	verify.mydomain.com -> “this is plain text”	Typically to verify ownership of a certain domain.
MX	mydomain.ch -> mailhost.mymailhoster.ch	Specify the mailserver for a domain.
NS	web.mydomain.ch -> ns1-123.azure-dns.com	Delegation of the web subdomain to a different DNS server.

DNS in the cloud

- No cloud works without DNS.
- (Almost) all cloud services are set up by default via a domain managed by a cloud provider (e.g., storage account).
 - Public DNS = DNS servers accessible from the internet (own or cloud provider domains).
 - Private DNS = DNS servers accessible from the VPC (own domains in your own network).
 - Split DNS = FQDNs (or entire domains) can be resolved publicly and privately, but receive different IPs – usually in corporate networks.
- Cloud providers offer corresponding services.
 - Azure DNS
 - AWS Route 53

DNS in cloud networks

- VMs and services in the VPC/VNET use cloud provider-managed DNS servers by default
- Custom DNS server
 - Managed DNS service
 - Self-managed DNS server (e.g., bind9) on VMs or similar



<https://learn.microsoft.com/en-us/azure/dns/private-dns-scenarios>

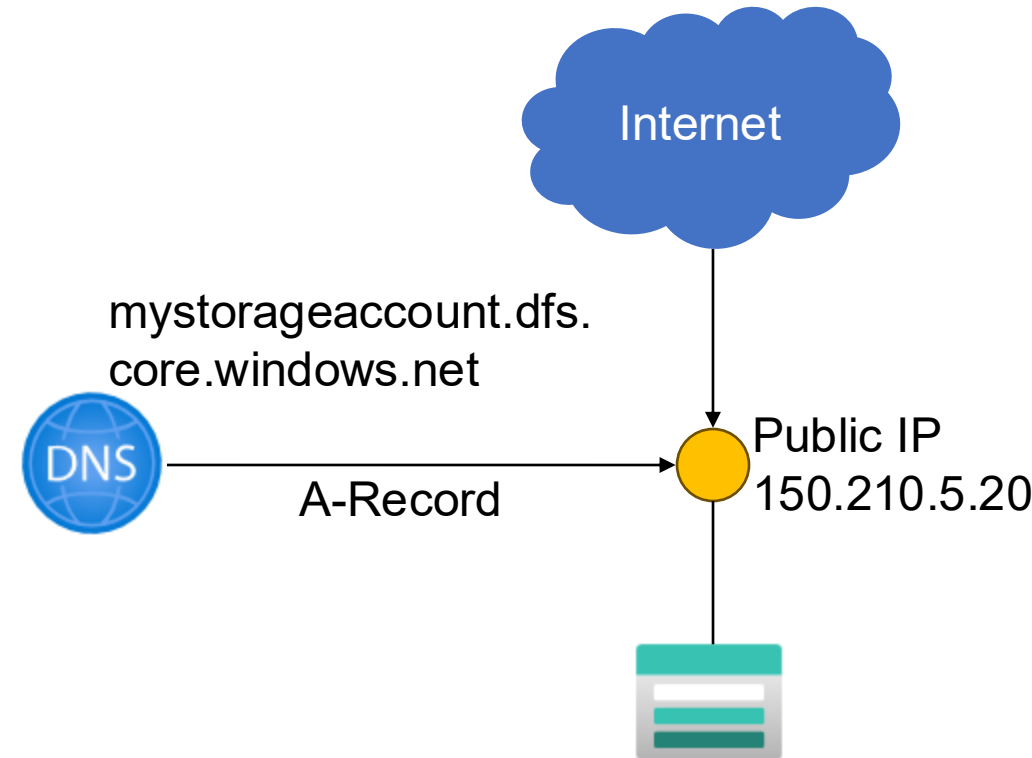
Private network integration of cloud services

Network integration for cloud services

- Cloud services (e.g., storage) can be accessed via the following paths
 - Public IPs/DNS
 - Public IPs/DNS via hyperscaler backbone
 - Private IPs/DNS
- There are different concepts for inbound and outbound traffic to/from cloud services!

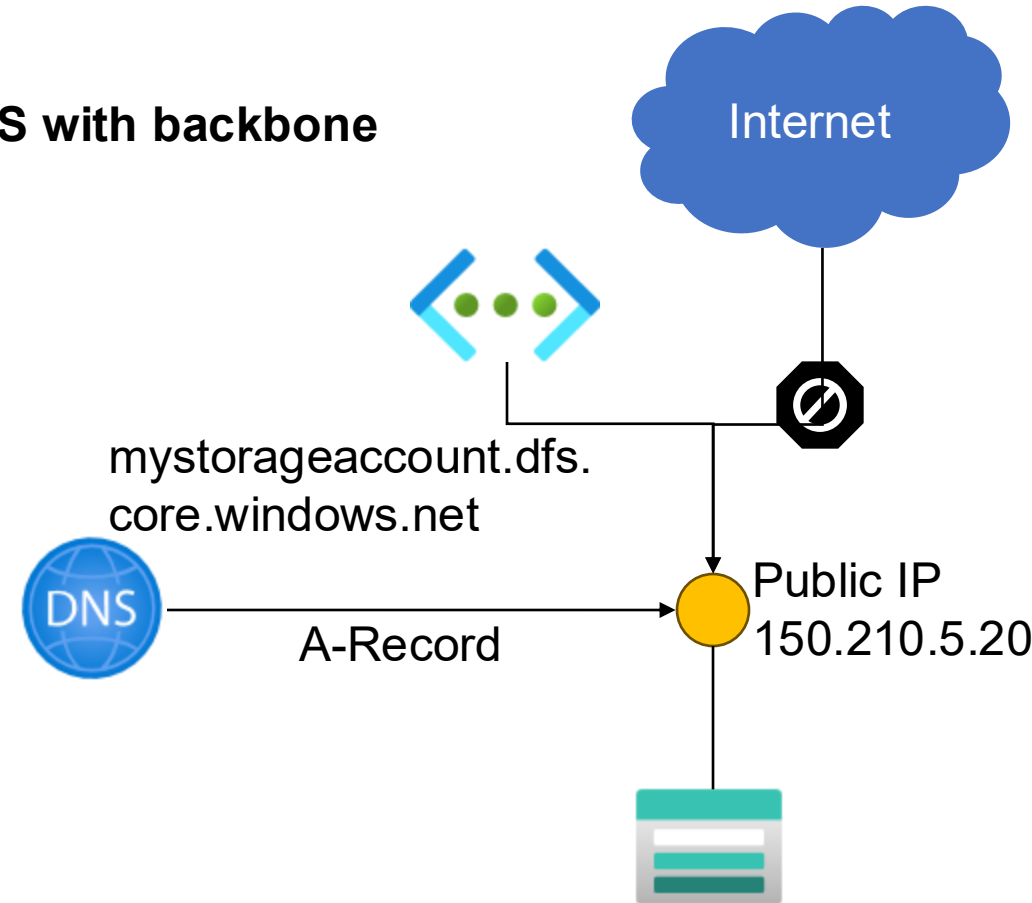
Network integration for cloud services – public IPs and DNS

- Cloud services can be accessed via a public IP address and domain, e.g.,
<https://mystorageaccount.dfs.core.windows.net/>
- IP restrictions (e.g., known IP addresses) are possible.
- Authentication/authorization can be configured depending on the service.
- Example: Publicly accessible S3 buckets.



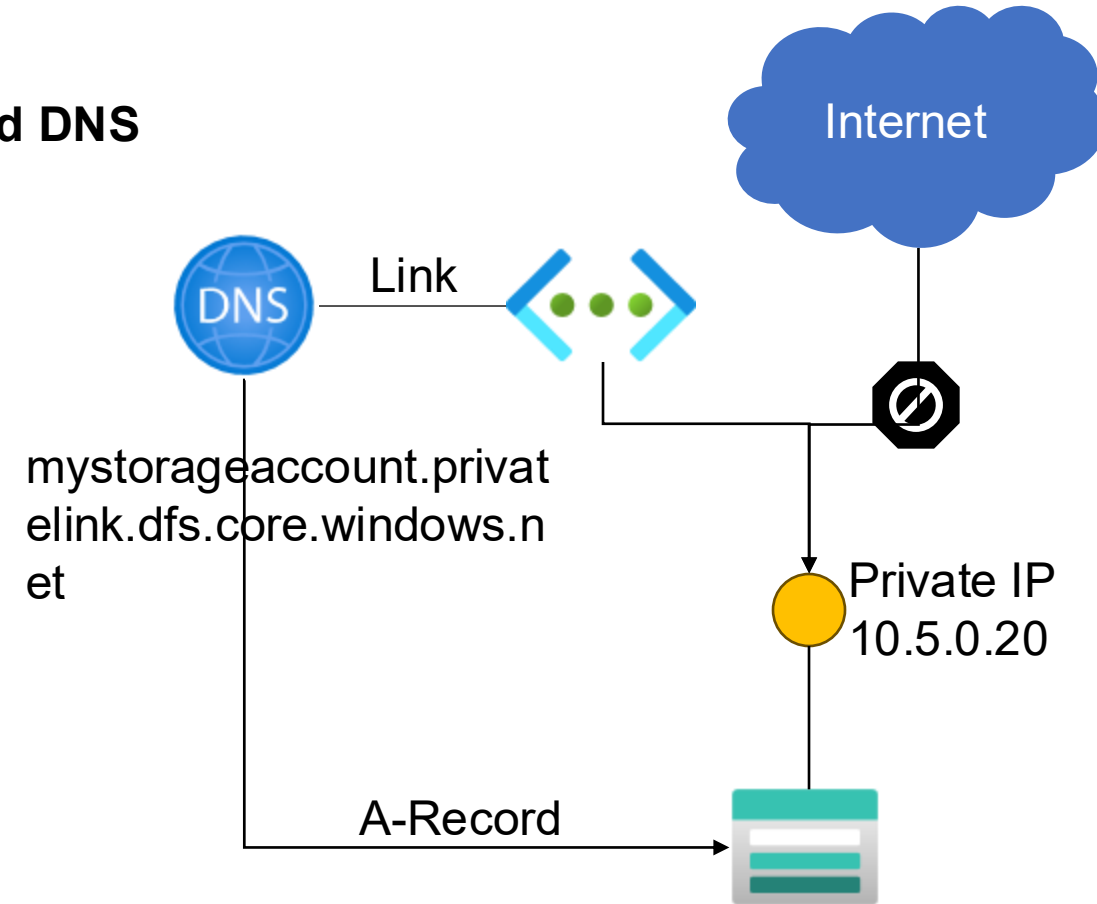
Network integration for cloud services – public IPs and DNS with backbone

- Cloud services are accessible via a public IP address and domain, e.g.,
<https://mystorageaccount.dfs.core.windows.net/>
- Cloud services are integrated into a VPC/VNET via SDN and can only receive traffic from there. Traffic flows via the cloud provider backbone.
- Example: Azure service endpoints or AWS VPC endpoint.



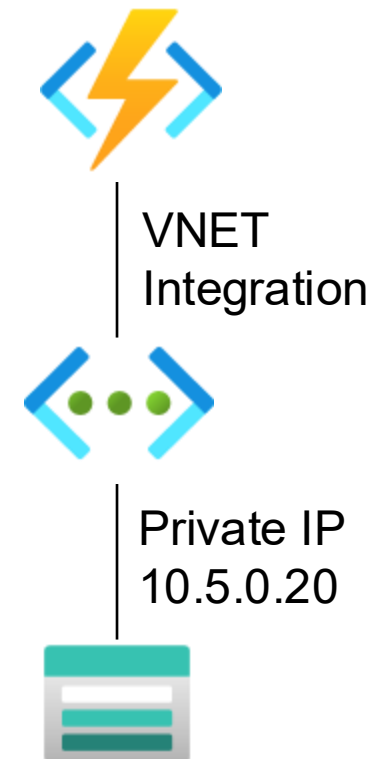
Network integration for cloud services – private IPs and DNS

- Cloud services can be accessed via a private IP and domain from a VPC or VNET. The domain is usually equivalent to the public setup.
- Cloud services are integrated into a VPC/VNET via an endpoint (private IP address) and can only receive traffic from there. Traffic flows purely through the VPC/VNET.
- Example: Azure Private Endpoints or AWS VPC Endpoints



Network integration for cloud services – Cloud service to VPC/VNET

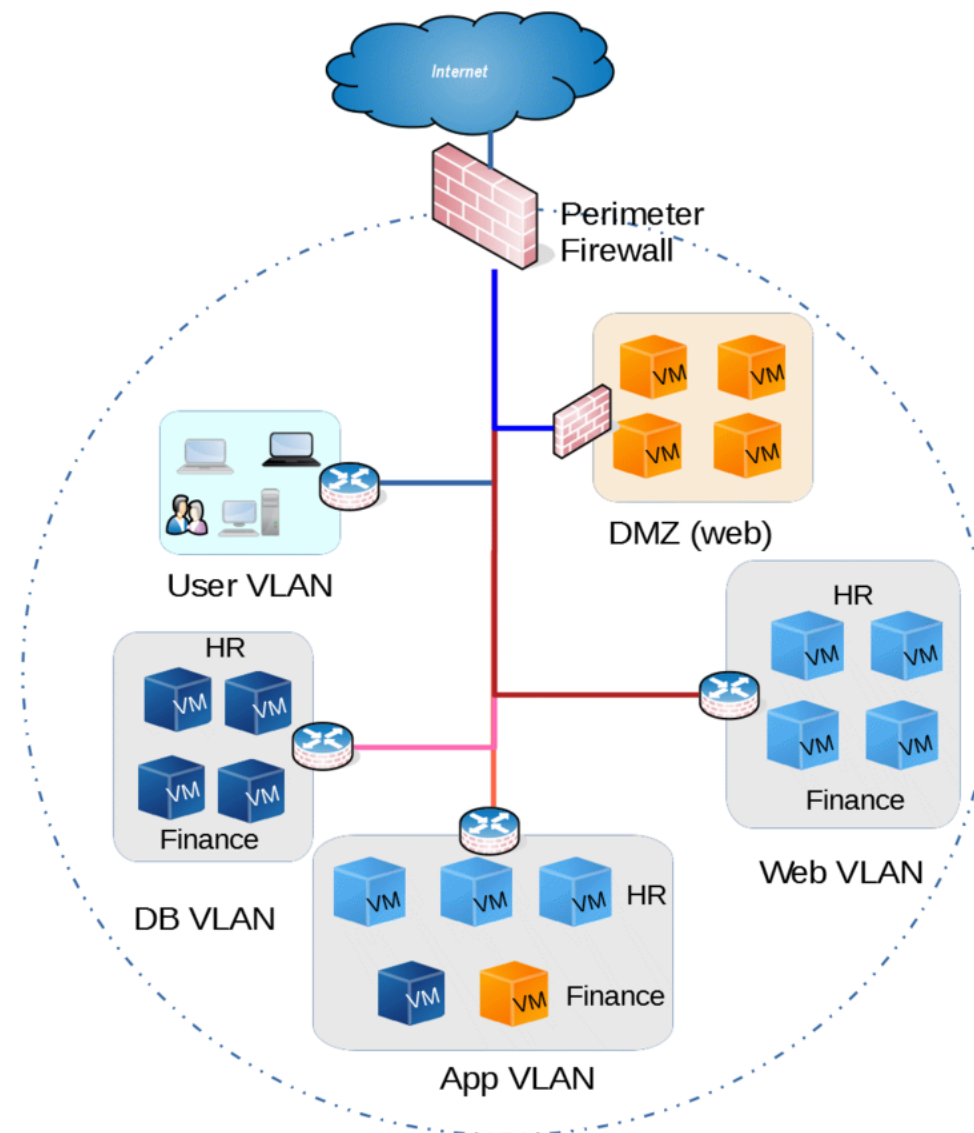
- Use case: Function must access a storage account configured with a private endpoint.
- PaaS cannot access VPC/VNET private IP addresses by default. IaaS uses simple NICs.
- Solution: Network integration of PaaS in VPC/VNET.
- Examples: Azure VNET integration or AWS Private Link.



Network segmentation

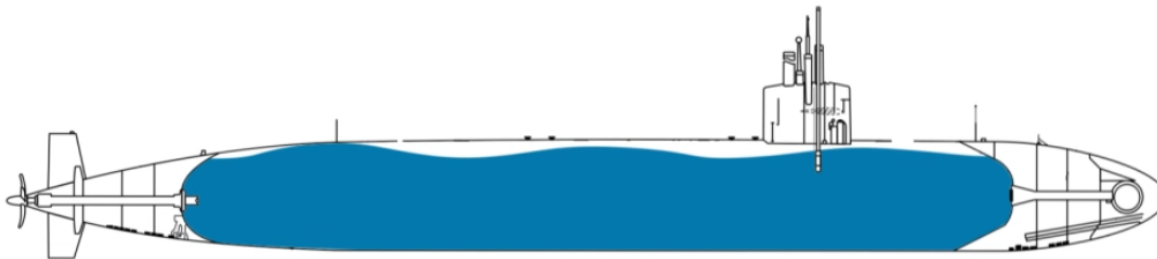
Networksegmentation

- Networks are divided
 - Based on user cases, rights, confidentiality
 - Rules between networks are clearly defined
 - Each zone has its own firewall
 - Clearly structured processes for approvals between different zones
- Agile mindset?
- Product teams VS “technology” zones?
- Real benefits?
- Firewall as a monolith

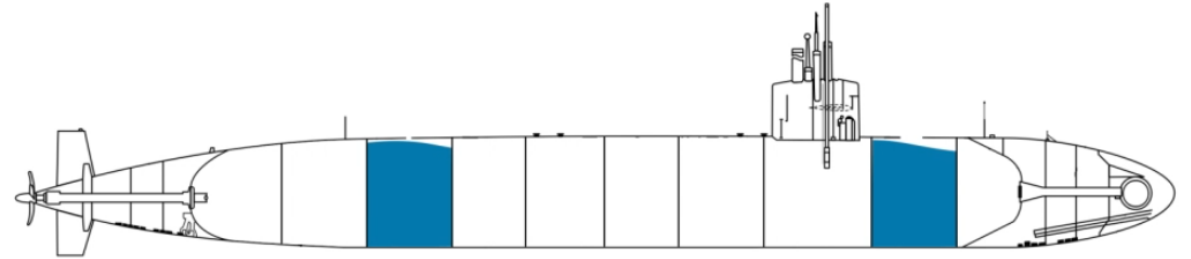


Microsegmentation

Without Micro-Segmentation

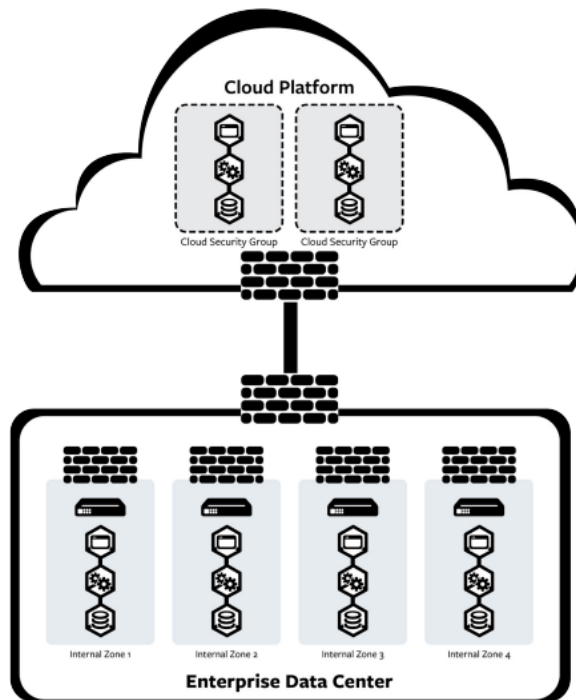


With Micro-Segmentation

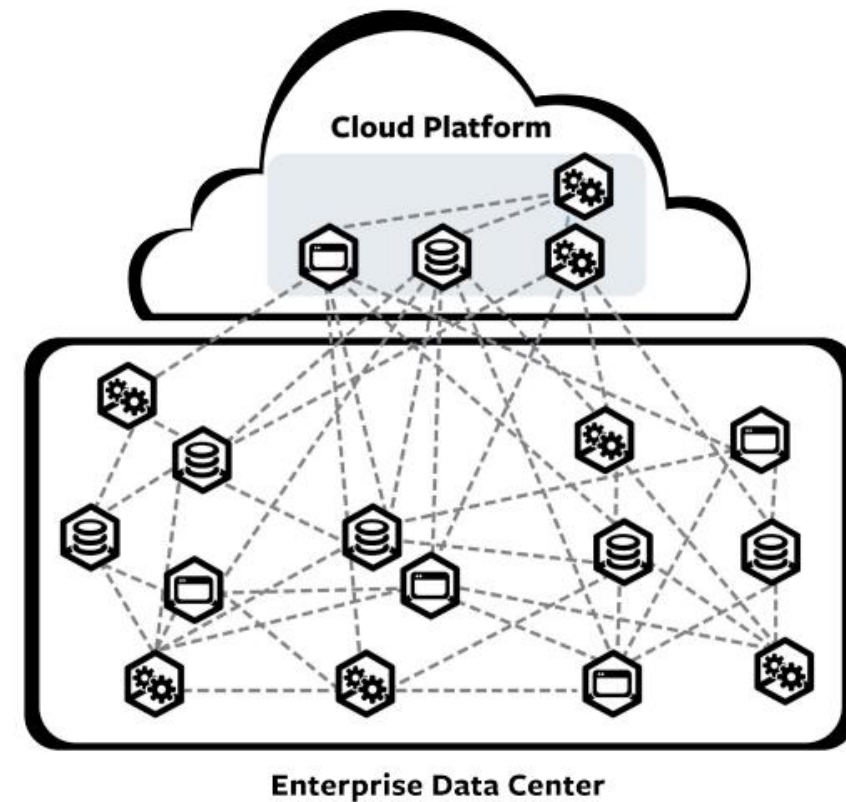


Microsegmentation

Datacenter-view



App-view



Potential communication paths
(not all shown) -----

Microsegmentation, how?

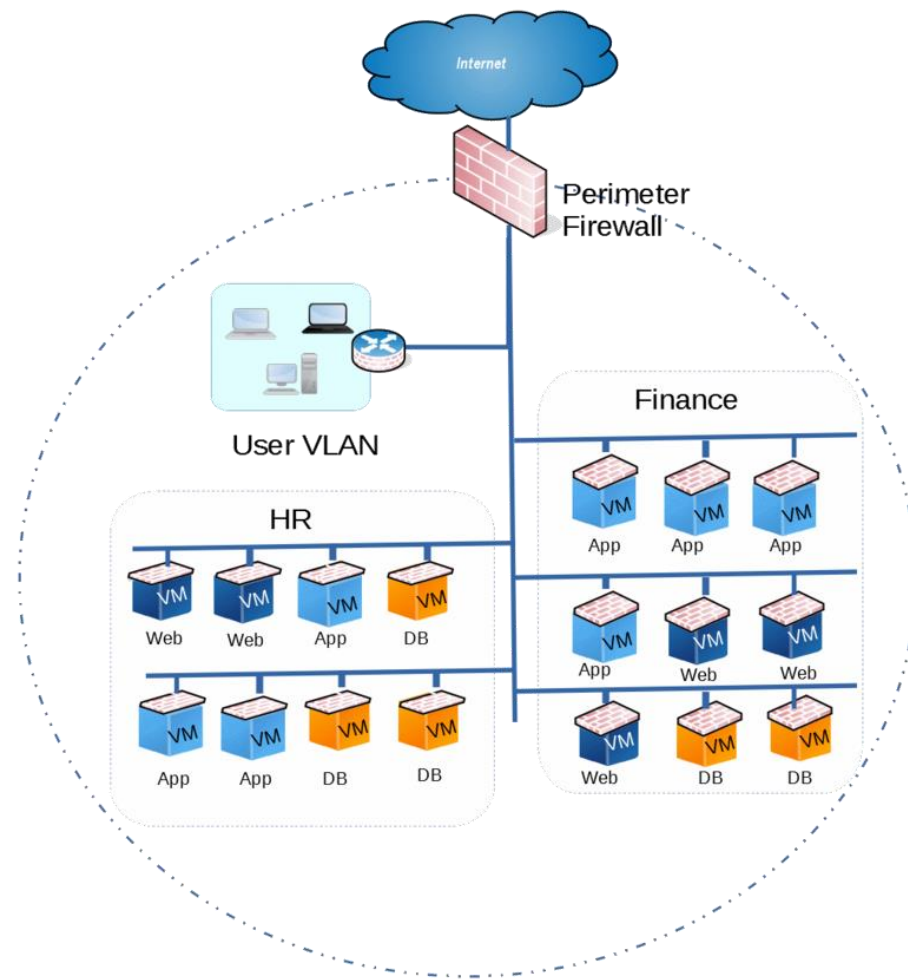
- Agent-based: Agent/software on infrastructure that controls access
- Network-based: Controls in network infrastructure (switches, SDNs, load balancers)
- Native cloud: Security groups/ACLs in network access for cloud providers/container platforms

Microsegmentation, why?

- Small attack surface in the network
- Containment of successful attacks
- Stronger compliance in the system
- Transparent management of policies
 - Central enforcement of policies
 - Decentralized management of policies

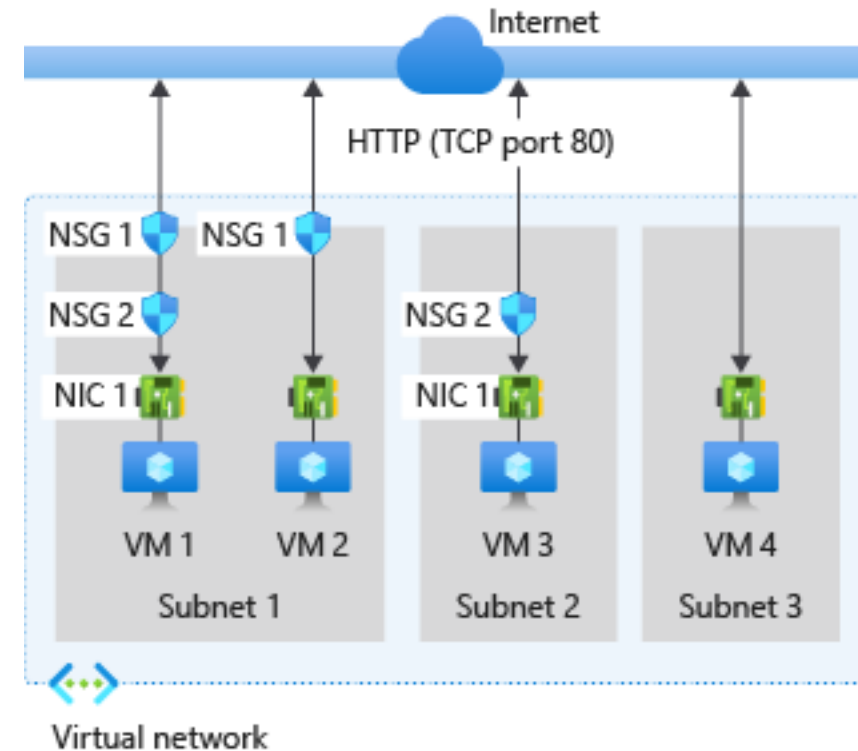
Networksegmentation

- Each subnet/machine has its own firewall.
- Not on the machine itself, but in the infrastructure.
- Virtual firewall/rule set for fixed port releases.
- Do not trust anybody.
- IaC to the rescue.
- Container platforms support this natively (more).
 - Cilium.



Networksegmentation in Azure

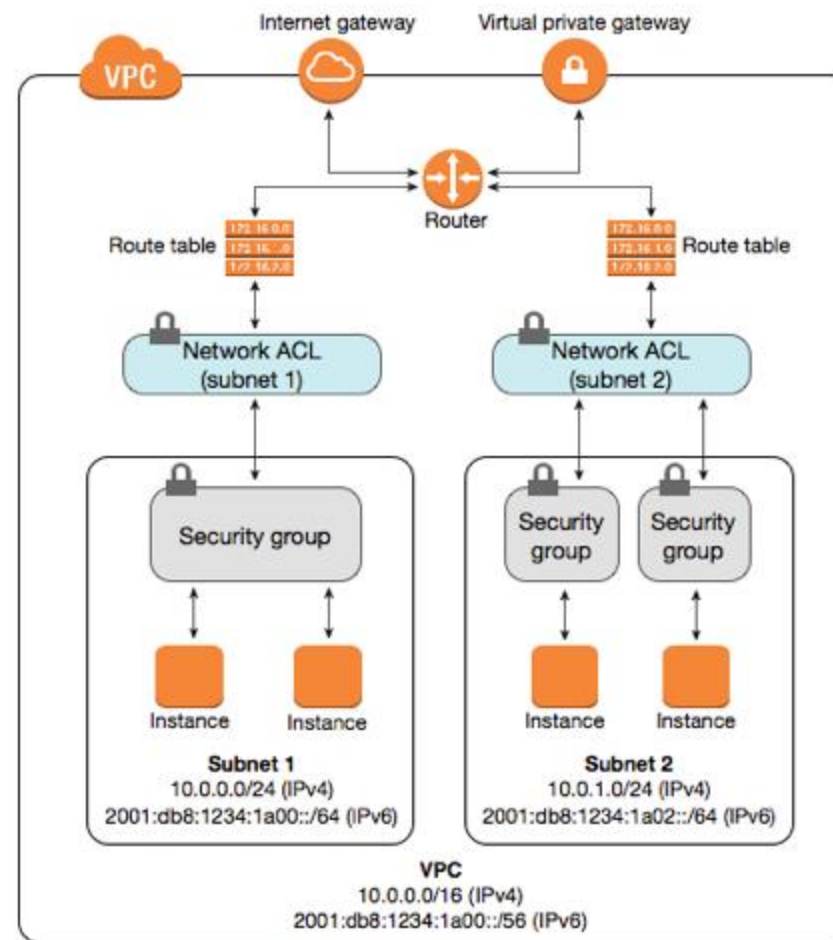
- Network Security Groups (NSGs) are L4 firewalls at the infrastructure level, applied to NICs or subnets.
- Standard rules allow traffic to and from the VNET, for example, but otherwise block all other traffic from outside.
- Further development Application Security Groups (ASGs)
 - Grouping of applications and their mapping, e.g., to VMs
 - Allows flexible scaling without adjustments, as it works on “applications” and not IPs
 - Only works with applications in Azure



<https://learn.microsoft.com/en-us/azure/virtual-network/network-security-group-how-it-works>

Networksegmentation in AWS

- Traffic is forwarded via the routing instances
- Network ACLs protect access to the subnet (VPC)
- In addition, each instance can be assigned one or more security groups.



Network pricing

Network Pricing

- Cost factors
 - Outgoing traffic (VPC -> Internet) per GB
 - Peering traffic (VPC <-> VPC) – always costs inbound and outbound per GB
 - Geographic destination of outgoing traffic
 - Gateways (NAT, VPN, etc.)
 - Public IPv4 / IPv6 addresses

- Incoming traffic is usually free of charge. However, depending on the service, costs may still be incurred (e.g., write access during upload).

Data Transfer	Price
Data Transfer In	Free
Data transfer within same Availability Zone	Free
Data transfer from Azure origin to Azure CDN*	Free
Data transfer from Azure origin to Azure Front Door Standard/Premium	Free

Intra-continental Data Transfer	Price
Between regions within North America Between regions within Europe	CHF 0.016 per GB
Between regions within Asia Between regions within Oceania Between regions within Middle East and Africa	CHF 0.064 per GB
Between regions within South America	CHF 0.128 per GB

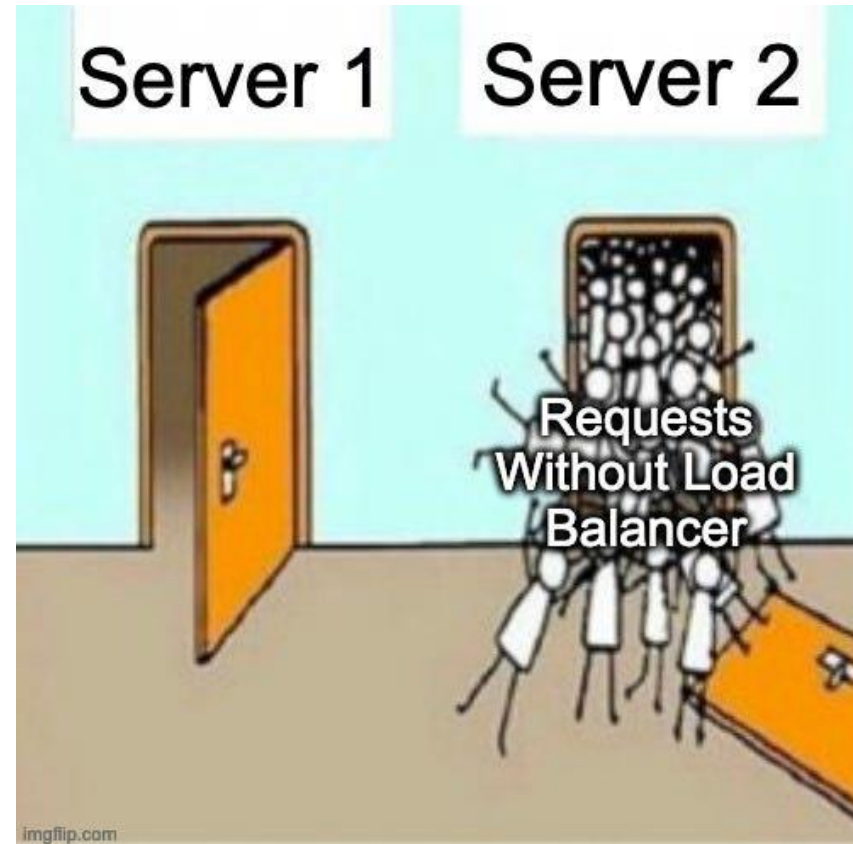
<https://azure.microsoft.com/en-us/pricing/details/bandwidth/>

Networking - Summary

- Clouds offer logical networks for hosting applications in a private IP address range
- PaaS services can integrate into these logical networks (inbound and outbound traffic)
- Network segmentation is designed for microsegmentation, although zoning is still possible



Load balancing motivation



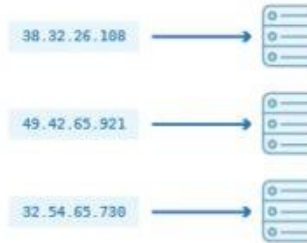
Load balancing concepts

The Five Most Common Load Balancing Methods

static

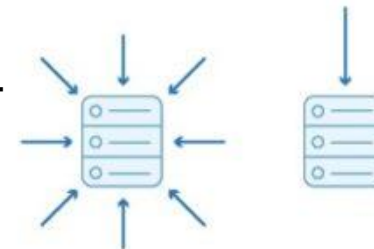


Round Robin



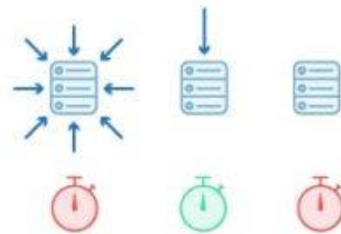
IP Hash

dynamic



Least Connections

dynamic



Least Response Time



Least Bandwidth

Load balancing – dynamic algorithms

- **Least connection:** All servers have the same number of resources; the server with the fewest connections gets the next connection.
- **Weighted least connection:** Servers have different resources; the server with the least weighted connections gets the next connection.
- **Weighted response time:** Calculates the average response time and gives the server with the lowest response time the next connection.
- **Resource-based:** Gives the server with the freest resources (CPU/RAM) the next connection.

Load balancing – static algorithms

- **Round robin:** Distributes connections based on multiple A/AAAA entries in the DNS.
- **Weighted round robin:** Implements a weighted round robin based on weights assigned to the DNS entries.
- **IP hash:** Combines the source and destination IP addresses and calculates a hash based on them. A specific server is requested based on this hash.

Load balancing L3/4 vs L7

TCP/IP Model		OSI Model		
Application Layer	HTTP, HTTPS, SMTP, IMAP, FTP, DNS, NNTP	Application		} 'Layer 7' Load Balancing
		Presentation		
		Session		
Transport	UDP, TCP, SCTP	Transport		} 'Layer 4' Load Balancing
Internet		Network		
Network Access (Link)		Data Link		
		Physical		

Load balancing L3/4 vs L7

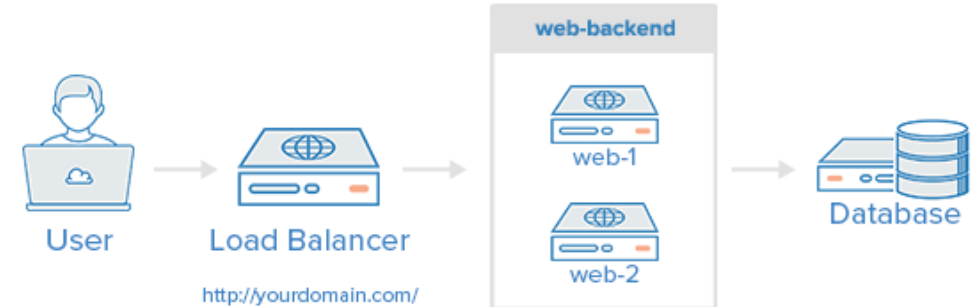
Layer 4:

- Fast
- Protocol-independent

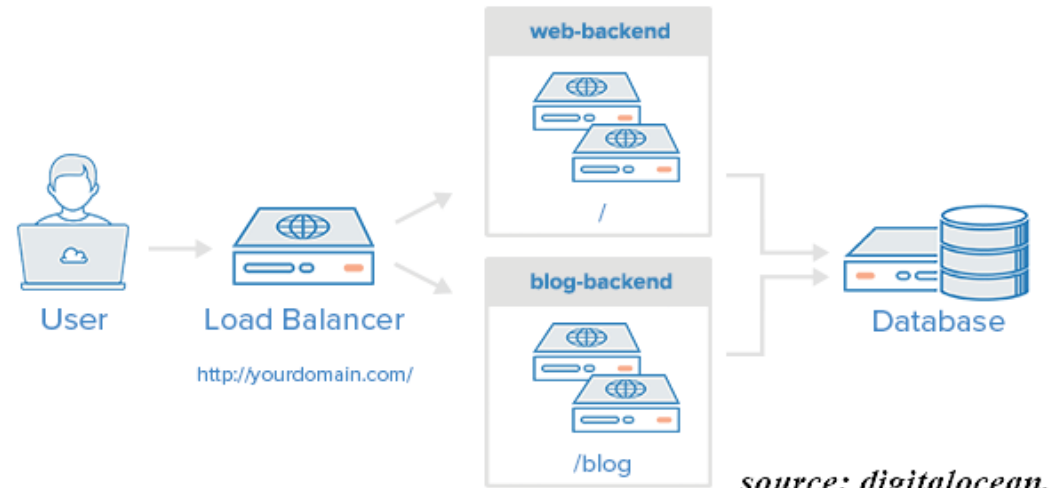
Layer 7

- Request awareness
- Session awareness
- HTTPS termination
- WAF

Layer 4 Load Balancing



Layer 7 Load Balancing



source: digitalocean.com

Load balancing services

AWS load balancing

Feature	Application Load Balancer	Network Load Balancer	Gateway Load Balancer	Classic Load Balancer
Load Balancer type	Layer 7	Layer 4	Layer 3 Gateway + Layer 4 Load Balancing	Layer 4/7
Target type	IP, Instance, Lambda	IP, Instance, Application Load Balancer	IP, Instance	-
Terminates flow/proxy behavior	Yes	Yes	No	Yes
Protocol listeners	HTTP, HTTPS, gRPC	TCP, UDP, TLS	IP	TCP, SSL/TLS, HTTP, HTTPS
Reachable via	VIP	VIP	Route table entry	-

<https://aws.amazon.com/de/elasticloadbalancing/features/?nc=sn&loc=2&dn=1>

Azure load balancing

Load balancing

[Give feedback](#)

Overview

Help me choose

Service comparison

Tutorial

Load Balancing Services

- Application Gateway
- Front Door
- Load Balancer
- Traffic Manager

Does your application require Layer 7 load balancing capability?

Application Gateway

- Internal and public configurations
- Regional layer 7 load balancer
- SSL/TLS offloading

[Show more](#)

Front Door

- Global layer 7 load balancer
- Site acceleration
- SSL/TLS offloading

[Show more](#)

Load Balancer

- Layer 4 load balancing
- Internal and public configurations
- High availability across zones

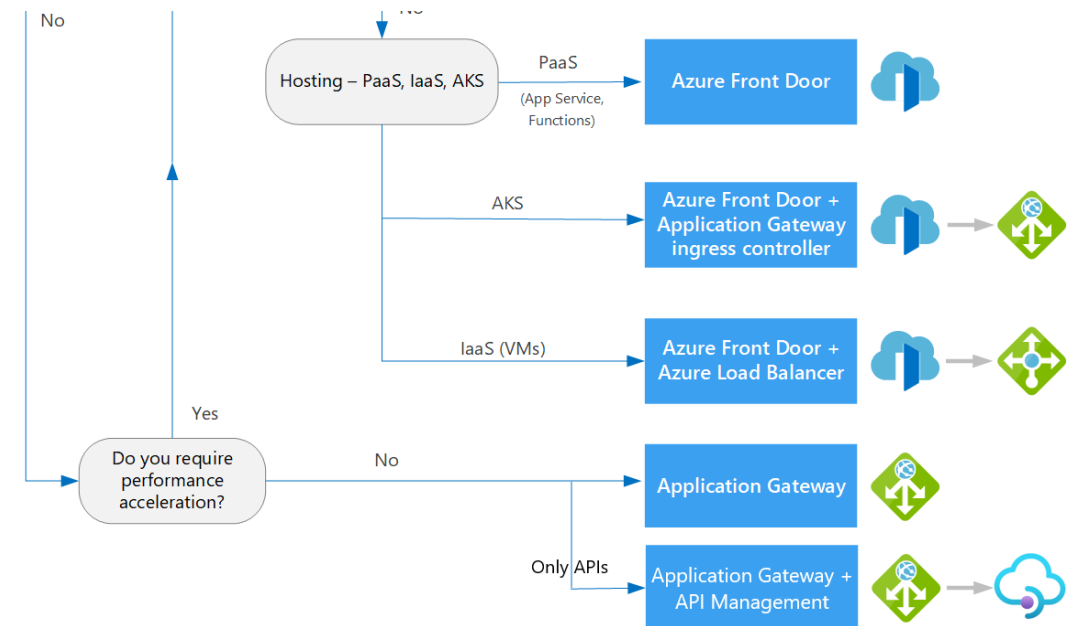
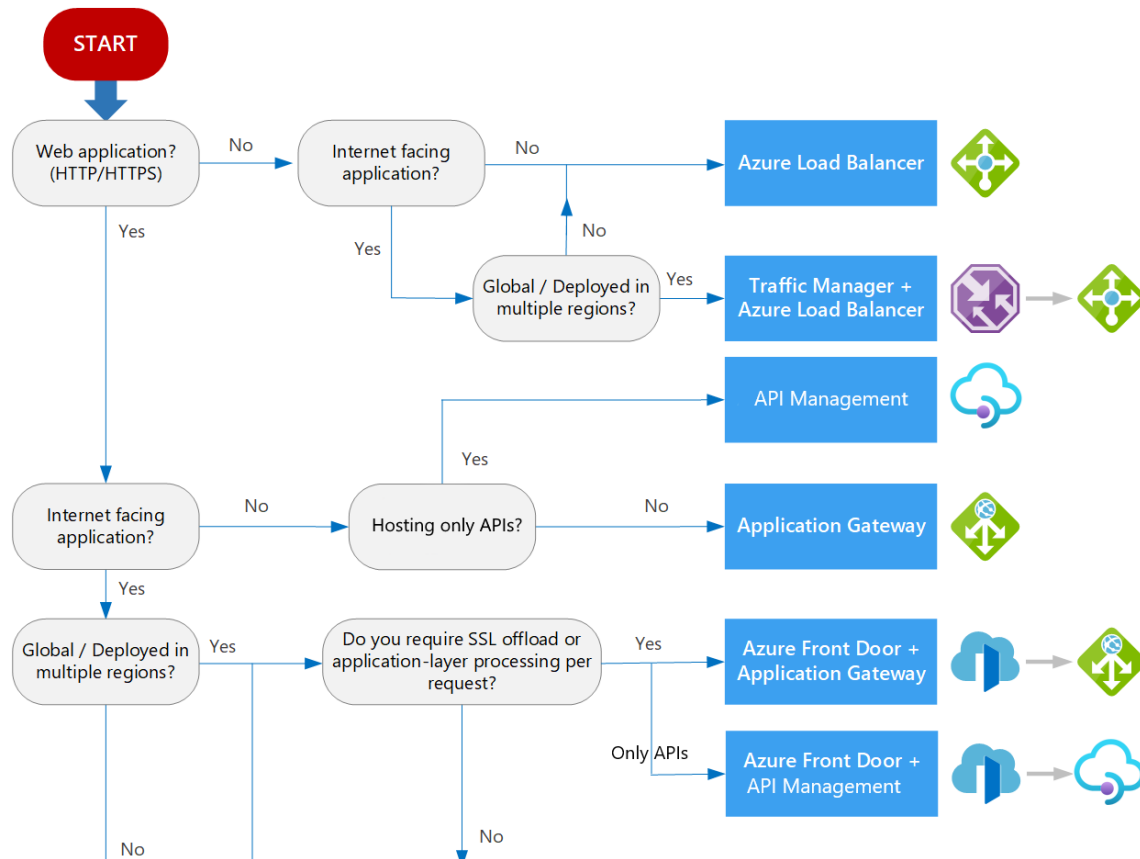
[Show more](#)

Traffic Manager

- DNS-based traffic load balancer
- Global across Azure regions
- High availability

[Show more](#)

Azure load balancing



<https://learn.microsoft.com/en-us/azure/architecture/guide/technology-choices/load-balancing-overview>

Load balancing use cases

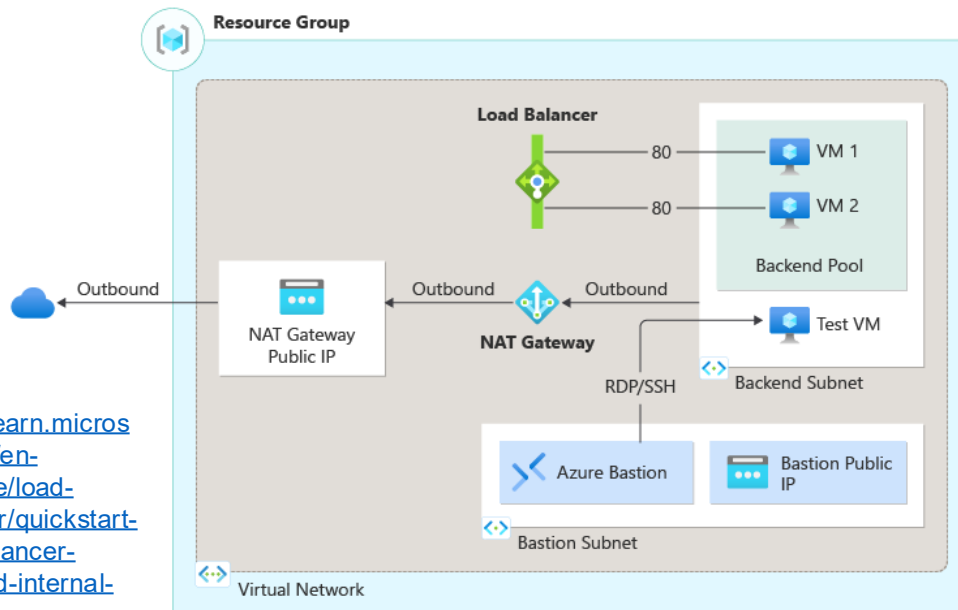
Load balancing – private und public use cases

Private load balancing

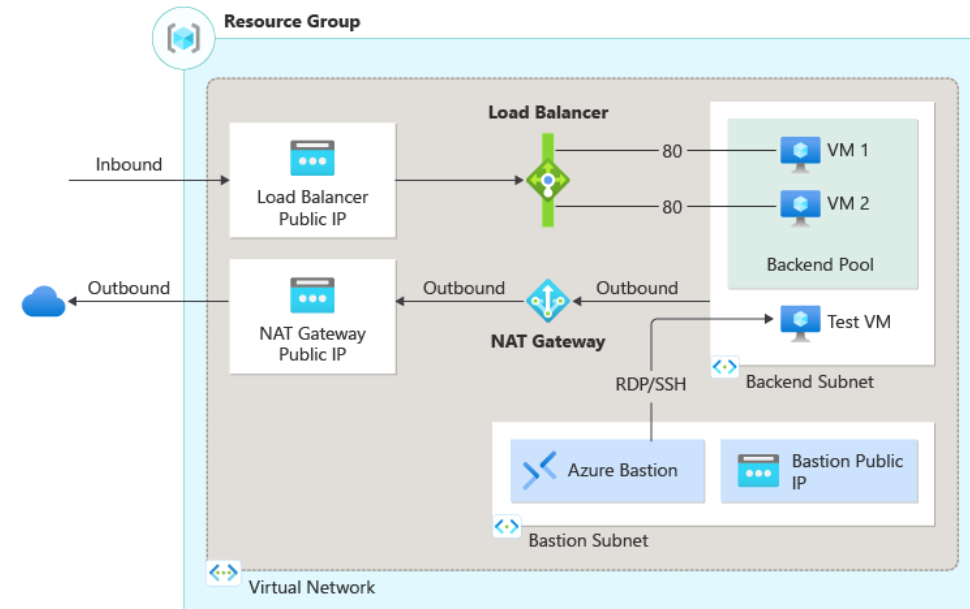
- Load balancing inside a private network for private clients (i.e. service to service)
- Clients are inside the network

Public load balancing

- Load balancing from a public endpoint (i.e. IP address) to private or public endpoints
- Clients are outside the network



<https://learn.microsoft.com/en-us/azure/load-balancer/quickstart-load-balancer-standard-internal-portal>

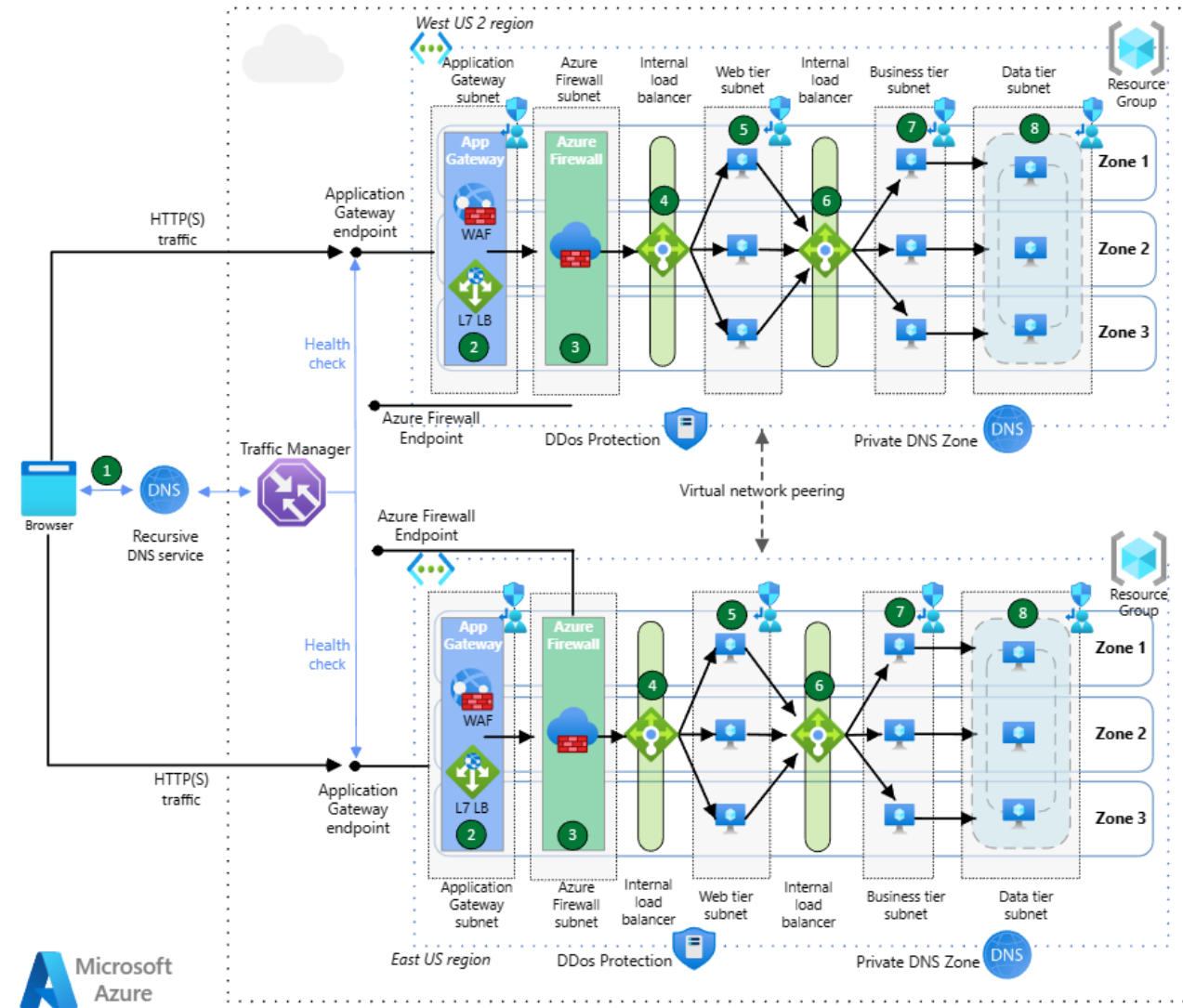


<https://learn.microsoft.com/en-us/azure/load-balancer/quickstart-load-balancer-standard-public-portal>

Load balancing – Global vs Regional

Global load balancing

- The endpoint is accessible on the WWW, traffic is routed to the client at the nearest possible edge location in the cloud provider network and from there on to the service
- Use case: Global load balancing to optimize latency and/or cross-regional availability

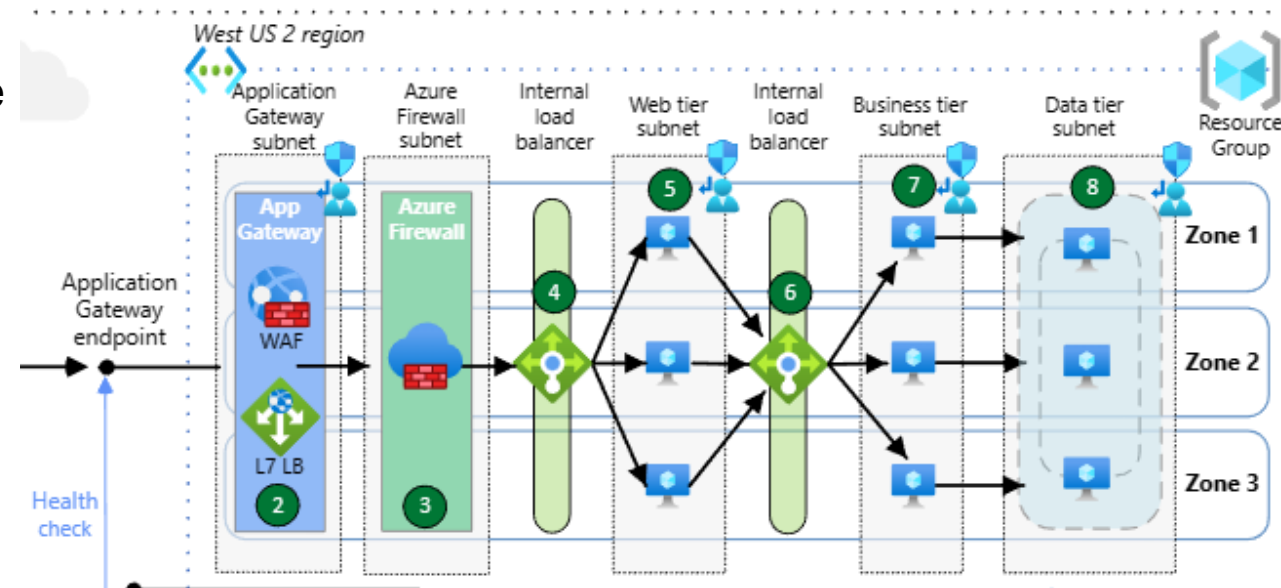


<https://learn.microsoft.com/en-us/azure/architecture/high-availability/reference-architecture-traffic-manager-application-gateway>

Load balancing – Global vs Regional

Regional load balancing

- The endpoint of the load balancer is accessible from the WWW but physically tied to a region.
- The backends of the load balancer are in the same region (but can also be global).
- Use case: Load distribution in the region for high availability via availability zones, for example.
- L3/4 and L7 load balancers supported.



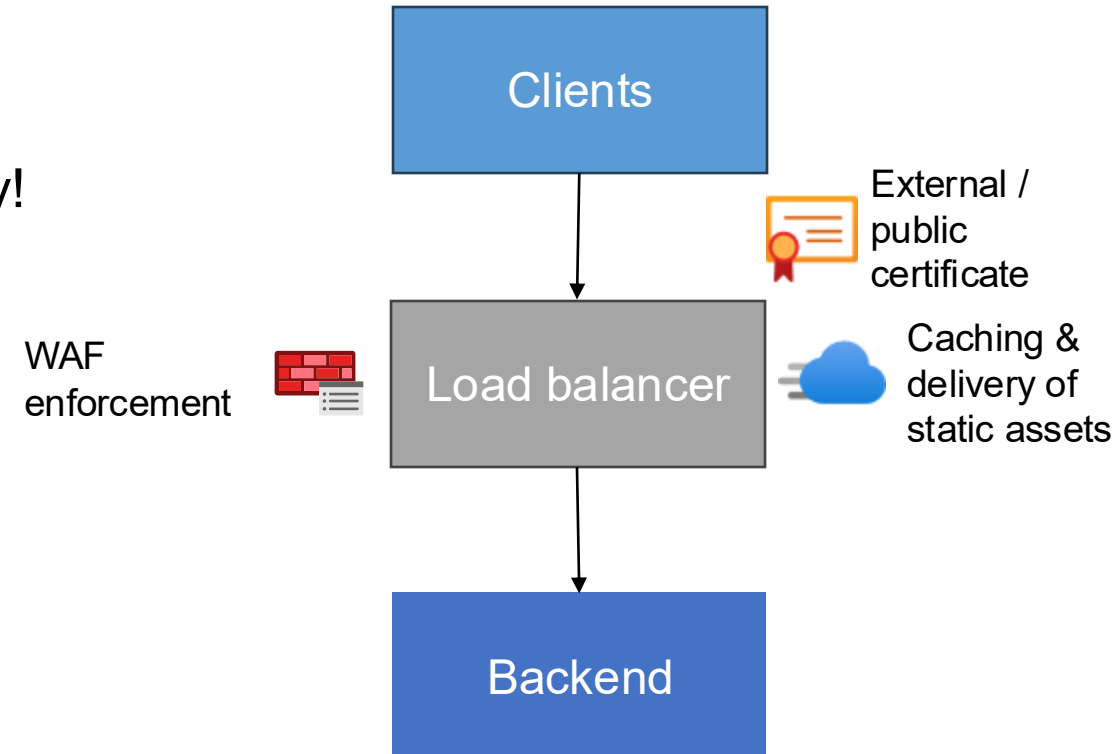
<https://learn.microsoft.com/en-us/azure/architecture/guide/technology-choices/load-balancing-overview>

Load balancing additional features

Load balancing

Only applicable if the load balancer is a reverse proxy!

- TLS termination
- Web Application Firewall (WAF)
- Content Delivery Network (CDN)



Load balancing pricing

Load Balancing Pricing

- L4 Load Balancer
 - Number of rules per hour
 - Per GB of data

- L7 Load Balancer (e.g., Application Gateway)
 - Prices and features vary depending on SKU (e.g., with or without WAF)
 - Fixed costs (public IP, number of instances)
 - Per capacity unit per hour (2,500 connections, 1 GB per hour)

	Application Gateway Basic
Fixed	CHF 0.0216 per gateway-hour
Capacity Unit ¹	CHF 0.0064 per capacity unit-hour

Application Gateway Standard	Web Application Firewall
CHF 0.2157 per gateway-hour	CHF 0.3913 per gateway-hour
CHF 0.0088 per capacity unit-hour	CHF 0.0152 per capacity unit-hour

<https://azure.microsoft.com/en-us/pricing/details/load-balancer/> und <https://azure.microsoft.com/en-us/pricing/details/application-gateway/>

Load Balancing – Summary

- L3/4 load balancers are protocol-agnostic and fast.
- L7 load balancers are protocol-aware and can offer additional features such as TLS termination, session-based load balancing, etc.
- Load balancing can be performed using various algorithms and metrics (round robin, least connections, etc.).
- Load balancing can be static or dynamic.



Homework

- Review the material from this lesson.
- Complete [Assignment 12](#).
- Create your own VNET with the following learning path:
 - [Introduction to Azure Virtual Networks](#)

